

ARM PrimeCell™

Advanced Audio CODEC Interface (PL041)

Technical Reference Manual



ARM PrimeCell Advanced Audio CODEC Interface (PL041)

Technical Reference Manual

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Release information

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Preface

This preface introduces the ARM PrimeCell Advanced Audio CODEC Interface (PL041) and its reference documentation. It contains the following sections:

- *About this document* on page vi
- *Further reading* on page ix
- *Feedback* on page x.

About this document

This document is a technical reference manual for the ARM PrimeCell AACI (PL041).

Intended audience

This document has been written for hardware and software engineers implementing System-on-Chip designs. It provides the necessary information to enable designers to integrate the peripheral into a target system as quickly as possible.

Using this manual

This document is organized into the following chapters:

Chapter 1 *Chapter 1 Introduction*

Read this chapter for a brief introduction to the PrimeCell AACI (PL041). It lists the features of the PrimeCell AACI and includes a simplified block diagram.

Chapter 2 *Chapter 2 Functional Overview*

Read this chapter for a description of the AC-link and the major functional blocks of the PrimeCell AACI.

Chapter 3 *Chapter 3 Programmer's Model*

Read this chapter for a description of the PrimeCell AACI registers and programming details.

Chapter 4 *Programmer's Model for Test*

Read this chapter for a description of the logic in the PrimeCell AACI for functional verification and production testing.

Appendix A *Appendix A ARM PrimeCell AACI (PL041) Signal Descriptions*

Read this appendix for details of the PrimeCell AACI signals.

Appendix B *Frame format*

Read this appendix for details on the frame format of the **AACISDATAOUT** and **AACISDATAIN** signals.

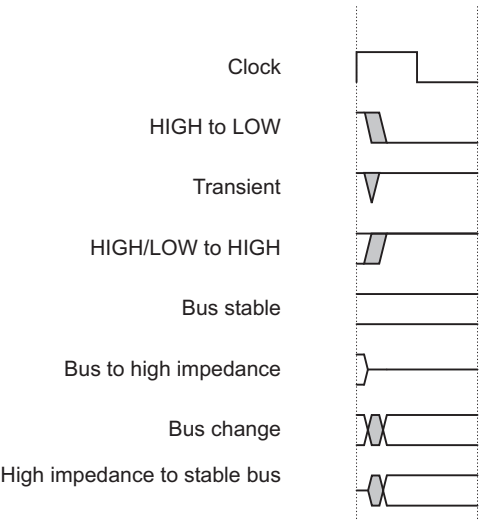
Typographical conventions

The following typographical conventions are used in this document:

bold	Highlights ARM processor signal names, and interface elements such as menu names. Also used for terms in descriptive lists, where appropriate.
<i>italic</i>	Highlights special terminology, cross-references and citations.
typewriter	Denotes text that can be entered at the keyboard, such as commands, file names and program names, and source code.
<u>typewriter</u>	Denotes a permitted abbreviation for a command or option. The underlined text can be entered instead of the full command or option name.
<i>typewriter italic</i>	Denotes arguments to commands or functions where the argument is to be replaced by a specific value.
typewriter bold	Denotes language keywords when used outside example code.

Timing diagram conventions

This manual contains one or more timing diagrams. The following key explains the components used in these diagrams. Any variations are clearly labeled when they occur. Therefore, no additional meaning should be attached unless specifically stated.



Key to timing diagram conventions

Shaded bus and signal areas are undefined, so the bus or signal can assume any value within the shaded area at that time. The actual level is unimportant and does not affect normal operation.

Further reading

This section lists publications by ARM Limited, and by third parties.

ARM periodically provides updates and corrections to its documentation. See <http://www.arm.com> for current errata sheets and addenda.

See also the ARM Frequently Asked Questions list at:
<http://www.arm.com/DevSupp/Sales+Support/faq.html>

ARM publications

This document contains information that is specific to the ARM PrimeCell AACI (PL041). Refer to the following documents for other relevant information:

- *ARM Specification* (Rev 2.0) (ARM IHI 0011)
- *ARM PrimeCell Advanced Audio CODEC Interface (PL041) Integration Manual* (PL041 INTM 0000)
- *ARM PrimeCell Advanced Audio CODEC Interface (PL041) Design Manual* (PL041 DDES 0000).

Other publications

This section lists relevant documents published by third parties.

- *Cirrus Logic CS4299 device specification*
- *National Semiconductor LM4549 device specification*
- *Wolfson Microelectronics WM9704Q device specification.*

Feedback

ARM Limited welcomes feedback both on the ARM PrimeCell AACI (PL041), and on the documentation.

Feedback on the ARM PrimeCell AACI (PL041)

If you have any comments or suggestions about this product, please contact your supplier giving:

- the product name
- a concise explanation of your comments.

Feedback on this document

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- the document title
- the document number
- the page number(s) to which your comments refer
- a concise explanation of your comments.

General suggestions for additions and improvements are also welcome.

Chapter 1

Introduction

This chapter describes the ARM PrimeCell Advanced Audio CODEC Interface (PL041). It contains the following section:

- *About the ARM PrimeCell AACI (PL041)* on page 1-2.

1.1 About the ARM PrimeCell AACI (PL041)

The ARM PrimeCell *Advanced Audio CODEC Interface* (AACI) is an *Advanced Microcontroller Bus Architecture* (AMBA) slave block that connects to the *Advanced Peripheral Bus* (APB). The PrimeCell AACI provides communication to off-chip CODECs that support the AC-link protocol.

Examples of these CODECs include:

- the WM9704Q from Wolfson Microelectronics Ltd
- the CS4299 from Cirrus Logic
- and the LM4549 from National Semiconductor Corporation.

AC97 CODECs offer a route for high-quality audio, modem support, and provision for headsets.

1.1.1 Features of the PrimeCell AACI

The PrimeCell AACI has the following features:

- Four channels are available and can be in operation simultaneously. This allows support for four different sample rates simultaneously.
- The channels can be serviced either by *Direct Memory Access* (DMA) or by the processor.
- Each channel has transmit and receive FIFOs.
- The four channels allow independence between modem, audio, headset and microphone channels.

Figure 1-1 is a simplified block diagram of a PrimeCell AACI.

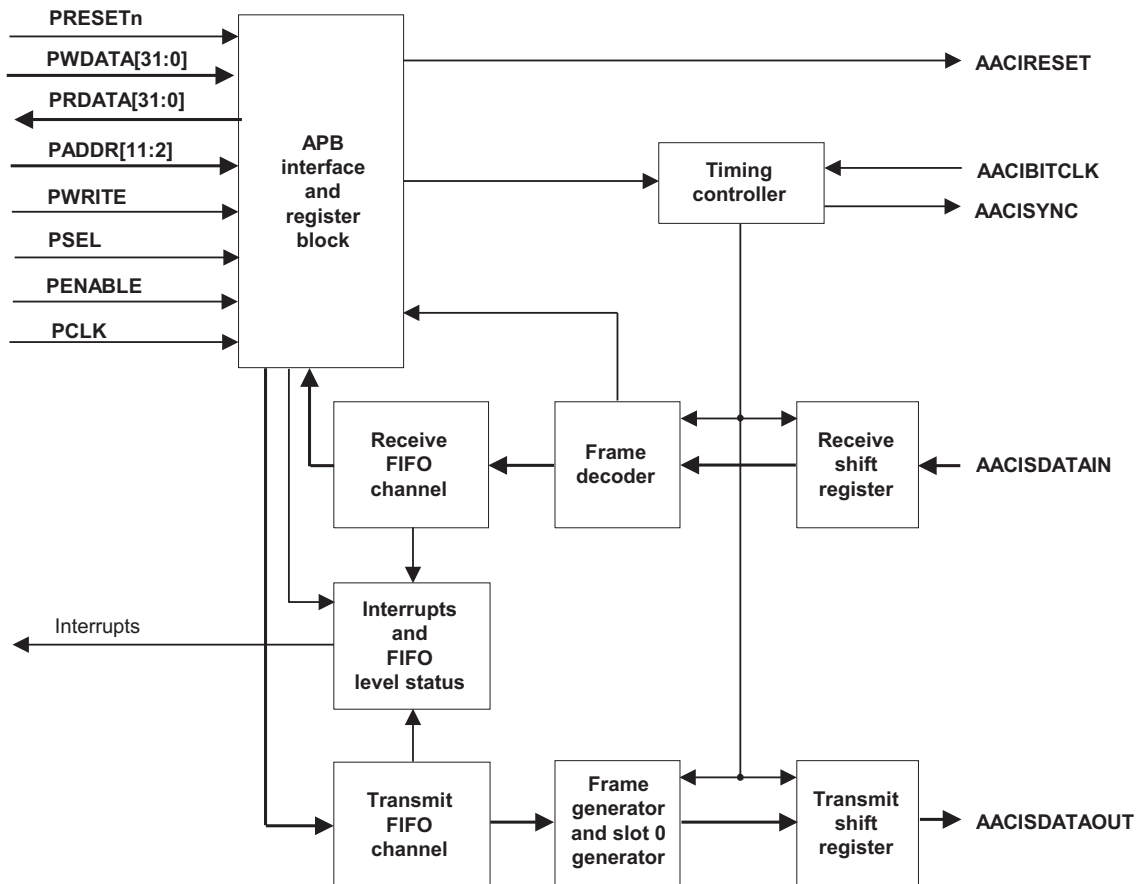


Figure 1-1 Simplified block diagram of a PrimeCell AACI showing one channel

Chapter 2

Functional Overview

This chapter provides an overview of the ARM PrimeCell AACI (PL041). It contains the following sections:

- *ARM PrimeCell AACI (PL041) overview* on page 2-2
- *AC-link description* on page 2-3
- *PrimeCell AACI functional description* on page 2-10
- *DMA interface bus protocol* on page 2-20.

2.1 ARM PrimeCell AACI (PL041) overview

The PrimeCell AACI provides communication to off-chip CODECs that support the AC-link protocol. The connection between the PrimeCell AACI and the external chip is known as an AC-link (for a detailed description of the AC-link see *AC-link description* on page 2-3). Figure 2-1 shows the PrimeCell AACI connected to a compliant CODEC.

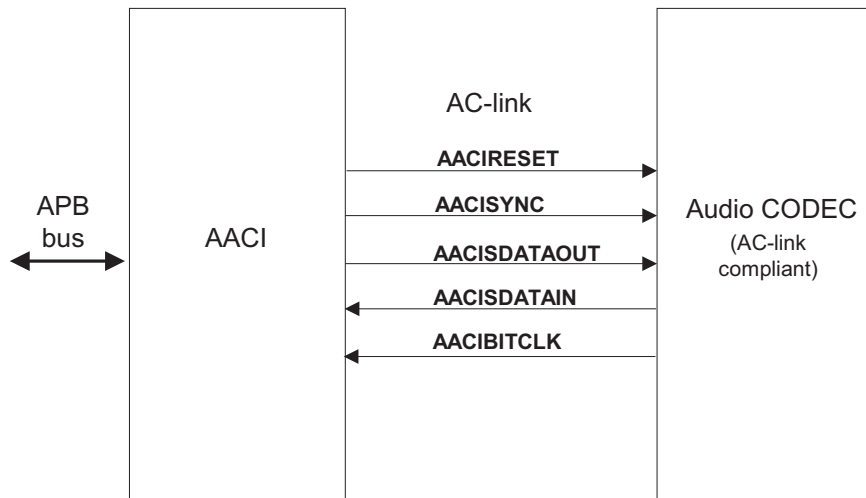


Figure 2-1 PrimeCell AACI connected to a compliant CODEC

The PrimeCell AACI contains logic that controls the AC-link to the audio CODEC and also controls the interface to the ARM *Advanced Peripheral Bus* (APB).

The PrimeCell AACI has four channels available. These can operate simultaneously, therefore allowing support for four different sample rates simultaneously. The channels can be serviced either by DMA or by the microprocessor.

2.2 AC-link description

The AC-link is a bidirectional, digital link between the PrimeCell AACI and a compliant CODEC. It uses a serial *Time Division Multiplexed* (TDM) format over five wires. The AC-link is a fixed rate, serial *Pulse Code Modulation* (PCM) digital stream. It handles multiple input, and output audio streams, in addition to control register accesses employing a TDM scheme. The AC-link architecture divides each audio frame into 12 outgoing and 12 incoming data streams, each with 20-bit sample resolution.

The AC-link is described in the following subsections:

- *AC-link serial interface protocol*
- *PrimeCell AACI output frame on page 2-4*
- *PrimeCell AACI input frame on page 2-5*
- *Timing controller and AACISYNC controller, AC-link reset modes on page 2-6*
- *AC-link low-power mode on page 2-8.*

2.2.1 AC-link serial interface protocol

Figure 2-2 shows the make-up of a bidirectional audio frame. The frame is made up of a tag and a data phase each with slot(s). Slot 0 is the tag phase and slots 1 to 12 are the data phase. The data phase slots are each allocated to a different function.

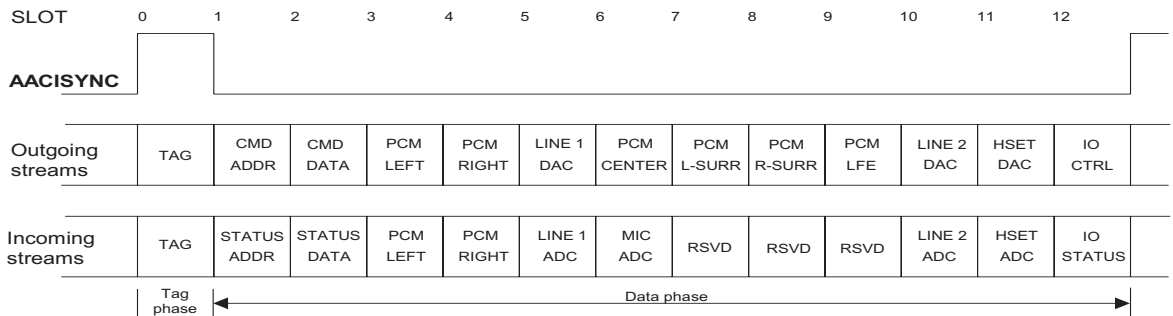


Figure 2-2 Bidirectional audio frame

Figure 2-3 on page 2-4 shows the key features of the AC-link frame:

- All signals are synchronized to **AACIBITCLK** that runs at 12.288MHz.
- The start of a frame is signaled by the rising edge of **AACISYNC**.
- The **AACISYNC** pulse is a regular occurrence and happens at the sample rate (for example 48kHz).

- The SCRA bits allow you to specify which CODEC the slot 1 and 2 data is for. It is possible to have up to four external CODECs connected to the AC-link with one CODEC being the primary CODEC and the others known as secondary CODECs. All other slot data is transmitted as normal and the external CODEC determines which slot they require.

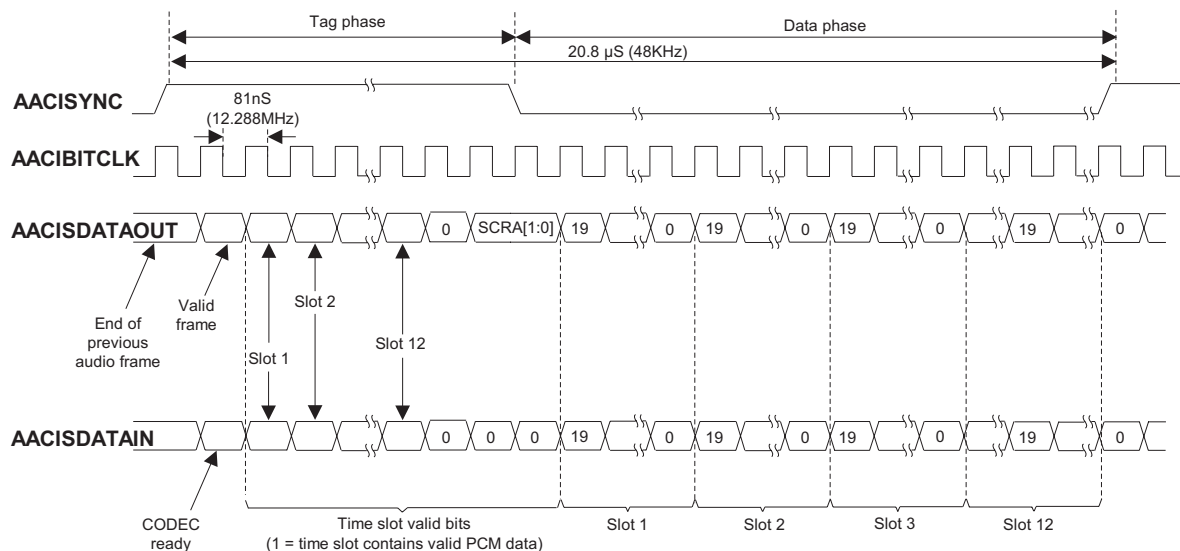


Figure 2-3 AC-link audio output frame

2.2.2 PrimeCell AACI output frame

The audio output frame (output from the interface) contains control and PCM data targeted for the external CODEC control registers and stereo *Digital-to-Analog Converter* (DAC). The tag slot, slot 0, contains 16 bits that tell the AC-link interface circuitry the validity of the following data slots.

A new audio frame is signaled with a LOW to HIGH transition of **AACISYNC**. **AACISYNC** is synchronous to the rising edge of the **AACIBITCLK**. **AACISYNC** changes from LOW to HIGH when the data on **AACISDATAOUT** is the final bit of the previous serial data frame. (The frame bit counter is equal to 255 and the rising edge of **AACIBITCLK**). On the next rising edge of **AACIBITCLK**, the interface drives **AACISDATAOUT** with the first bit of slot 0 (frame bit counter = 0). The external CODEC samples the **AACISDATAOUT** on the falling edge of **AACIBITCLK**. The PrimeCell AACI continues outputting the **AACISDATAOUT** stream on each successive rising edge of **AACIBITCLK**.

Figure 2-4 on page 2-5 is a timing diagram showing the start of an audio frame output.

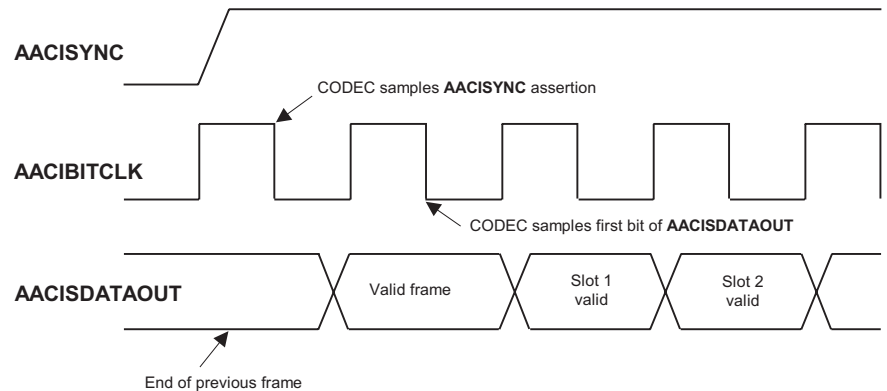


Figure 2-4 Start of audio frame output

The **AACISYNC** signal remains high for a total of 16 **AACIBITCLK**s at the beginning of each frame. (When the frame bit counter equals 15 and **AACIBITCLK** has a rising edge, **AACISYNC** returns to zero.)

Note

The frame format of **AACISDATAOUT** is shown in Table B-1 on page B-3.

2.2.3 PrimeCell AACI input frame

The input frame (input to the PrimeCell AACI) contains the status and PCM data from the external CODEC control registers and *Analog-to-Digital Converter* (ADC) data. The tag slot, slot 0, contains 16 bits that tell the PrimeCell AACI whether the external CODEC is ready and the validity of the data from the device subsections.

A new audio frame is signaled with a LOW to HIGH transition on **AACISYNC**. **AACISYNC** is synchronous to the rising edge of **AACIBITCLK**. On the next rising edge of **AACIBITCLK**, the CODEC drives **AACISDATAIN** with the first bit of slot 0. The PrimeCell AACI samples **AACISDATAIN** on the falling edge off **AACIBITCLK**. The external CODEC continues outputting the **AACISDATAIN** stream on each successive rising edge of **AACIBITCLK**. The external CODEC outputs data MSB first, in a MSB justified format. All reserved bits and slots are filled with zeros by the external CODEC.

Prior to any attempts at putting the external CODEC into operation the PrimeCell AACI must poll the first bit in the audio frame input (**AACISDATAIN** slot 0, bit 15) for an indication that the CODEC has gone into the CODEC READY state.

The first bit slot 0 receives is a global bit that flags whether the CODEC is in the CODEC READY state. If the CODEC READY is a 0, this indicates that the CODEC is not ready for normal operation. If the CODEC READY is a 1, control and status registers are in a fully operational state. You must then probe the power down control and status registers to determine which subsections, if any, are ready.

When the PrimeCell AACI has sampled CODEC READY, the next 12 bit positions sampled indicate which of the corresponding 12 time slots are assigned to input data streams, and contain valid data. Data is only stored in one of the FIFOs if AACIRXCR has its RX bit set. You must ensure that the FIFO control registers are set up to accept all received data.

On the rising edge of **AACIBITCLK**, before the last bit of the *tag slot*, **AACISYNC** is deasserted. From this point to the end of the frame the slots contain 20 bits of data. Figure 2-5 shows the start of an audio frame input.

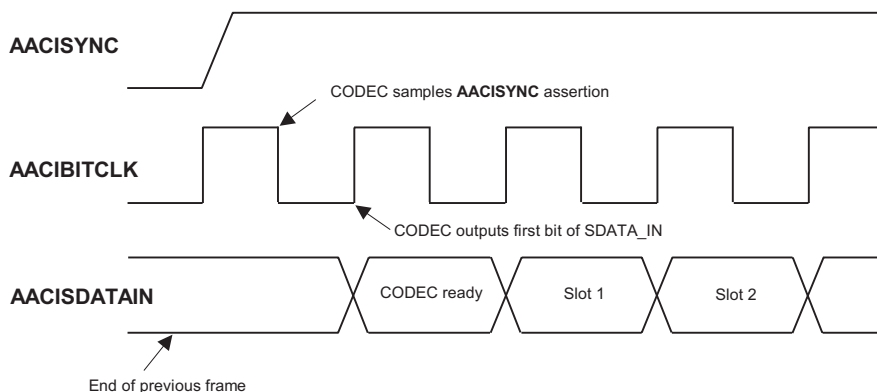


Figure 2-5 Start of audio frame input

Note

The frame format of **AACISDATAIN** is shown in Table B-2 on page B-5.

2.2.4 Timing controller and AACISYNC controller, AC-link reset modes

There are three methods to reset the external CODECs:

- *Cold reset* on page 2-7
- *Warm reset* on page 2-7
- *Register reset* on page 2-8.

The cold and warm resets are controlled through the AACISYNC and AACIRESET registers.

When a reset is asserted the external CODEC activates the **AACIBITCLK**. The **AACIBITCLK** is then used to generate the **AACISYNC** signal required for normal receive and transmit. An **AACIBITCLK** count is required to count to 255 before **AACISYNC** is reactivated for normal operation. A transmit on the **AACISDATAOUT** port does not occur until the software receives a CODEC ready.

Cold reset

A cold reset is achieved by asserting **AACIRESET** for the minimum specified time. A cold reset is required to restart the AC-link when bit PR5 is set in register 0x26 in the external CODEC. Figure 2-6 shows a timing diagram of the cold reset.

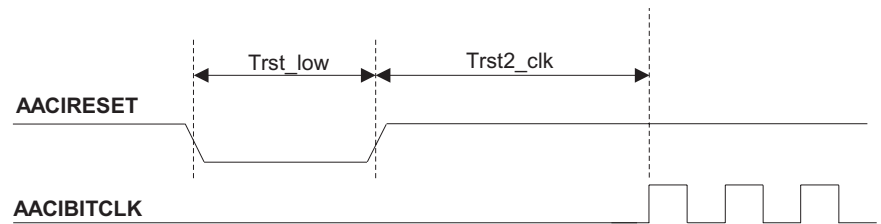


Figure 2-6 Cold reset timing diagram

The **Trst_low** time for all devices is a minimum of 1μs. The **Trst2_clk** time (**AACIRESET** inactive to **AACIBITCLK** startup delay) can vary depending on the external CODEC.

The **AACIRESET** pin is controlled through the AACIRESET register. This effectively puts the pin under software control.

Warm reset

A warm reset is required to restart the AC-link when bit PR4 is set in register (0x26) in the external CODEC. A warm reset allows you to reactivate the AC-link without losing information in the external CODEC registers. Figure 2-7 on page 2-8 shows a timing diagram of a warm reset.

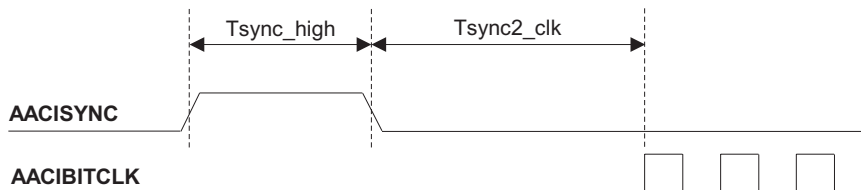


Figure 2-7 Warm reset timing diagram

The T_{sync_high} time is typically $1.3\mu s$ for most CODECs. The T_{sync2_clk} time (typically $162.8ns$) varies between devices. See the individual specifications for details. The **AACISYNC** signal has functionality that allows you to write to the **AACISYNC** register and force the **AACISYNC** pin to a desired value. This effectively puts the pin under software control for warm reset conditions.

When powered down, reactivation of the AC-link through reassertion of the **AACISYNC** signal must not occur for a minimum of four audio frame times ($4 \times 20.8\mu s$) following the frame in which the power down was triggered.

Register reset

The third reset mode provides a register reset to the external CODEC. The register reset allows all user accessible registers in the CODEC to be reset to their default, power-up values. A register reset occurs when any value is written to the CODEC register **00**.

2.2.5 AC-link low-power mode

The AC-link signals can be placed in low-power mode, when the power down control and status register (**0x26**) of the CODEC is programmed to the appropriate value, both **AACIBITCLK** and **AACISDATAIN** are brought to, and held at 0.

The controller drives **AACISYNC** and **AACISDATAOUT** LOW after the programming of the CODEC to this low-power, halted mode. This occurs after the end of slot 2 transmission. The **AACISL1TX** register is monitored by the controller to check if the address as indicated by bits 18:12 is **0x26**, and the **AACISL2TX** register bit 16 is monitored to see if it is set. Bit 16 in the **AACISL2TX** register relates to bit 12 (PR4) in the power down register address **0x26** of the external CODEC. This is so that the controller can recognize that the AC-link is to be brought into a low-power mode through the **AACISDATAOUT** port.

When the PrimeCell AACI is at the point where it is ready to program the external CODEC into its low-power mode, slots 1 and 2 are assumed to be the only valid stream in the output frame. It is assumed that all sources of audio input and output are neutralized or disabled.

2.3 PrimeCell AACI functional description

Figure 2-8 is a simplified block diagram of the PrimeCell AACI showing one of its four channels.

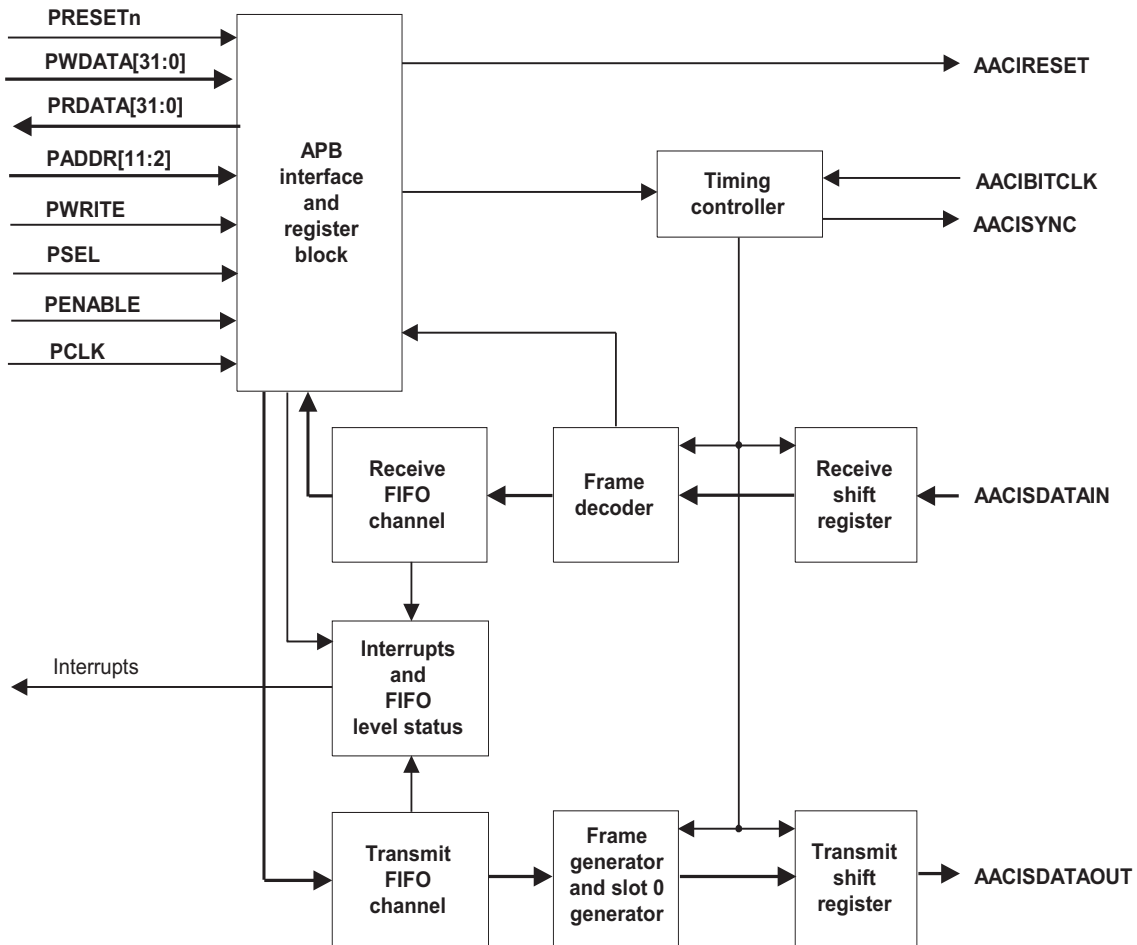


Figure 2-8 Simplified block diagram of a PrimeCell AACI showing one channel

The AACI consists of the following blocks:

- *AMBA APB interface and register block* on page 2-11
- *Receive FIFO channel* on page 2-11
- *Transmit FIFO channel* on page 2-11
- *Receive shift register* on page 2-11
- *Transmit shift register* on page 2-12

- *Frame generator and slot 0 generator* on page 2-12
- *Timing controller* on page 2-12
- *Interrupt and FIFO level status* on page 2-12.

2.3.1 AMBA APB interface and register block

The AMBA *Advanced Peripheral Bus* (APB) interface generates read and write decodes for accesses to status and control registers, and transmit and receive FIFO memories.

The AMBA APB interface is a local secondary bus that provides a low-power extension to the higher bandwidth AMBA *Advanced High-performance Bus* (AHB) or AMBA *Advanced System Bus* (ASB) within the AMBA system hierarchy. The AMBA APB groups narrow-bus peripherals to avoid loading the system bus and provides an interface using memory-mapped registers that are accessed under programmed control.

The register block stores data values written or to be read across the AMBA APB interface.

2.3.2 Receive FIFO channel

The receive FIFO is a 20-bit wide by eight deep memory buffer. Writes to the receive FIFO occur from the frame decoder section. Reads from the receive FIFO occur through the APB interface. The receive FIFO forms part of a channel and is described in *Channel description* on page 2-17.

2.3.3 Transmit FIFO channel

The transmit FIFO is a 20-bit wide by eight deep memory buffer. Writes to the transmit FIFO occur through the APB interface in the **PCLK** domain. Reads from the transmit FIFO occur through the frame generator section, which runs on **AACIBITCLK**. The transmit FIFO forms part of a channel and is described in *Channel description* on page 2-17.

2.3.4 Receive shift register

When the AC-link is active, the incoming data on **AACISDATAIN** is shifted into the receive shift register on the falling edge of **AACIBITCLK**. Data is clocked out of the shift register at the end of each slot.

2.3.5 Transmit shift register

When the AC-link is active, each slot data is clocked into the transmit shift register from the frame generator or slot 0 generator and is shifted out bit-by-bit on the rising edge of **AACIBITCLK**.

2.3.6 Frame generator and slot 0 generator

The slot 0 generator generates the slot 0 information and outputs it to the transmit shift register. The frame generator generates slots 1 to 12 and outputs them to the transmit shift register.

2.3.7 Frame decoder

The frame decoder qualifies the data of slots 1 to 12 that is output from the receive shift register, depending on data present in slot 0, and outputs it to the receive FIFO channel or slot receive registers.

2.3.8 Timing controller

When the AC-link is active, the timing controller drives the **AACISYNC** signal based on the **AACIBITCLK** signal from the off-chip CODEC. When the AC-link is inactive, the timing controller drives the internal register value on the **AACISYNC**.

2.3.9 Interrupt and FIFO level status

Individual maskable, active HIGH interrupts are generated by the PrimeCell AACI, and a combined interrupt output is also generated as an OR function of the individual interrupt requests.

2.3.10 Clock signals

The PrimeCell AACI uses two clock signals, the APB bus clock **PCLK** and **AACIBITCLK**. The maximum frequency of **PCLK** is 100MHz and the frequency of **AACIBITCLK** is fixed at 12.288MHz.

2.4 PrimeCell AACI operation

The operation of the PrimeCell AACI is described in the following sections:

- *Slot classification*
- *PrimeCell AACI channels* on page 2-14
- *Channel description* on page 2-17.

2.4.1 Slot classification

The PrimeCell AACI can support up to four different sample rates at a time. To allow the PrimeCell AACI to support all slots per frame, it is assumed that the sampling rate for the different types of data is the same. Slots in the same FIFO must have the same sample rates, different FIFOs can have different sample rates. For example, all audio data and all modem data is at the same sampling rate. Consider the case where the external CODEC supports the following channels:

- PCM left
- PCM right
- Modem1
- PCM playback centre
- PCM playback left surround
- PCM playback right surround
- PCM playback low frequency effects surround
- Modem 2
- Headset.

For this example you can program the transmit side of the PrimeCell AACI so that:

- audio data is in channel 1
- modem data is in channel 2
- headset data is in channel 3.

Because the PrimeCell AACI can also receive MIC data at a different rate this must be stored in channel 4.

The PrimeCell AACI is designed to allow you to store any slot data into any channel. If the external CODEC supports more than four sample rates you must decide which sample rates to allow.

2.4.2 PrimeCell AACI channels

The PrimeCell AACI has four channels and the functional blocks for these are described in *Channel description* on page 2-17. Each channel contains receive and transmit paths and associated control logic.

You can configure the control logic to allow the FIFOs to accept any data from or to any slot in a frame. The receive part of the channel is controlled through the AACIRXCR register and the transmit part of the channel is controlled through the AACITXCR register.

Receive data handling

Receive data handling through the AACIRXCR register controls the following:

- What slot data from the received frame is to be stored in the FIFO. The PrimeCell AACI does not store any other slots than those specified in these registers. You must ensure that all slot data stored in the FIFO are at the same sampling rate. Do not store more than eight slots in one receive FIFO as your system is required to service the FIFO more regularly.
- The length of time before a timeout interrupt is generated.
- If the FIFO is enabled.
- The number of bits in the slot that is required.
- If the channel can receive data.

Transmit data handling

Transmit data handling through the AACITXCR register controls the following:

- The slot in which FIFO data is to be transmitted. You must ensure that all the data in the FIFO is intended for slots with the same sampling rate.
- The number of data words that must be in the FIFO before the data within it can be transmitted. If the FIFO does not have enough data words and the CODEC has requested data an underflow interrupt is generated. The number of words in the FIFO must be equal to or greater than the number of slots that can be stored in the FIFO as indicated by the TX bits in the register. You must ensure that no more than eight slots are stored in the FIFO. If you try to set up AACITXCR to allow greater than eight slots in one transmit channel an underflow condition is signaled.
- The number of bits that need to be appended to the data from the CPU to make the word 20 bits.
- If the channel can transmit data.

If the external CODEC supports variable sample rates, you can program the CODEC to use them. If alternative rates are selected, the AC-link continues to run at 48KHz, but data is transferred across the link in bursts so that the desired sample rate is achieved. The external CODEC can demand samples at varying rates with the use of the data request disable bits in slot 1. The data request bits for all audio and modem data are expected to occur at the same time.

Slot determination for transmission

Slot 0 for transmission is determined by:

- the values in the AACITXCR register
- the data request bits
- the FIFO having valid data to send
- SCRA bits in the AACIMAINCR
- the slot registers with valid data to send.

If a slot does not have any data for transmission the PrimeCell AACI fills the slot with zeros. If the data request bits request data but the data is not available in the FIFO an underflow interrupt is generated. It is assumed that the CODEC sets the data request bits for a given sample rate all at the same time. For example, all the request bits for audio data are set at the same time. Only data with the same sample rate can be stored in the same FIFO and all data must be available for transmission when requested.

If the external CODEC does not support the data request disable bits or the variable rate extension, the bits are always zero meaning a sample rate of 48kHz. As slots 1 and 2 are always transmitted at 48kHz, the external CODEC does not supply data request disable bits for these.

Data transmission using registers

You can obtain data for transmission on slots 1, 2, and 12 from either of the channels or the registers AACISL1TX, AACISL2TX, and AACISL12TX. The recommended use of programming the external CODEC through slot 1 and slot 2 data is to use the channels during setup routines and the slot register at any other time. If you use the channels you must ensure that data is 20 bits in size and only one channel contains both slot 1 and slot 2 data. The data written into the slot 1, slot 2 and slot 12 registers is transmitted only when their corresponding S11TxEn, S12TxEn, and S112TxEn bits are set in the AACIMAINCR register. When you want to place the external CODEC into low-power mode you must use the AACISL1TX and AACISL2TX registers for data transmission as these have the logic required to set the low-power mode status bit.

When receiving the data for slots 1, 2, and 12 the data is stored in the channel and the AACISL1RX, AACISL2RX, and AACISL12RX registers if the appropriate enable bits are set. If the slot enable bits are not set then the data always goes to the registers.

When programming through slot registers you must ensure that slot 2 data is written before slot 1 data. The slot 2 data is required before slot 1 data as the PrimeCell AACI could start transmission before you have written to slot 2. If this happens the slot 2 data is not sent in this frame. If slots 1 and 2 are in separate channels, slot 2 must be stored in a channel on its own.

Frame generator and decoder

Figure 2-9 shows a block diagram of the logic to generate and decode the frames and the way channels interact with the generate and decode logic.

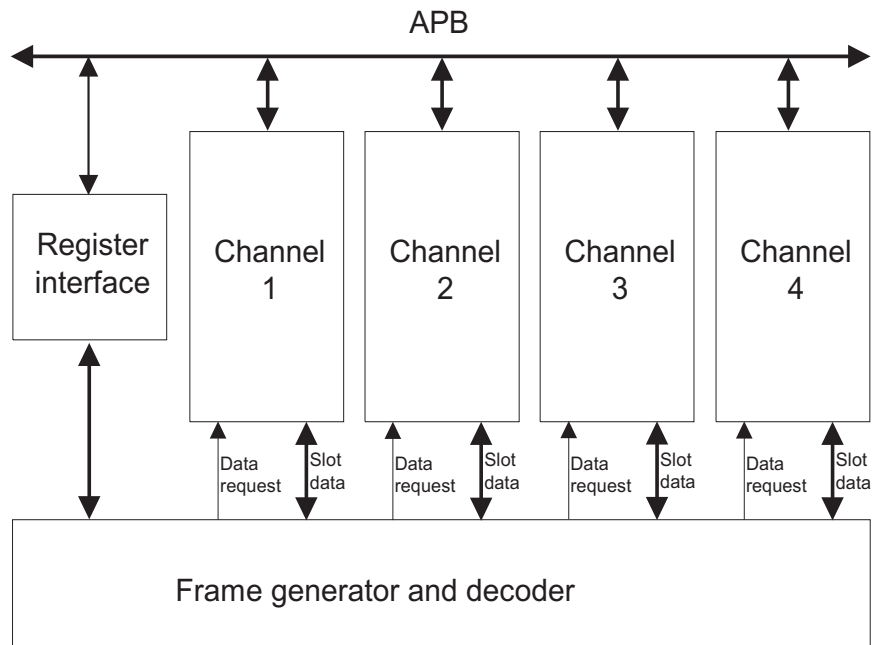


Figure 2-9 Frame generator and decoder

2.4.3 Channel description

All four channels of the PrimeCell AACI are identical. Figure 2-10 shows a block diagram of one channel.

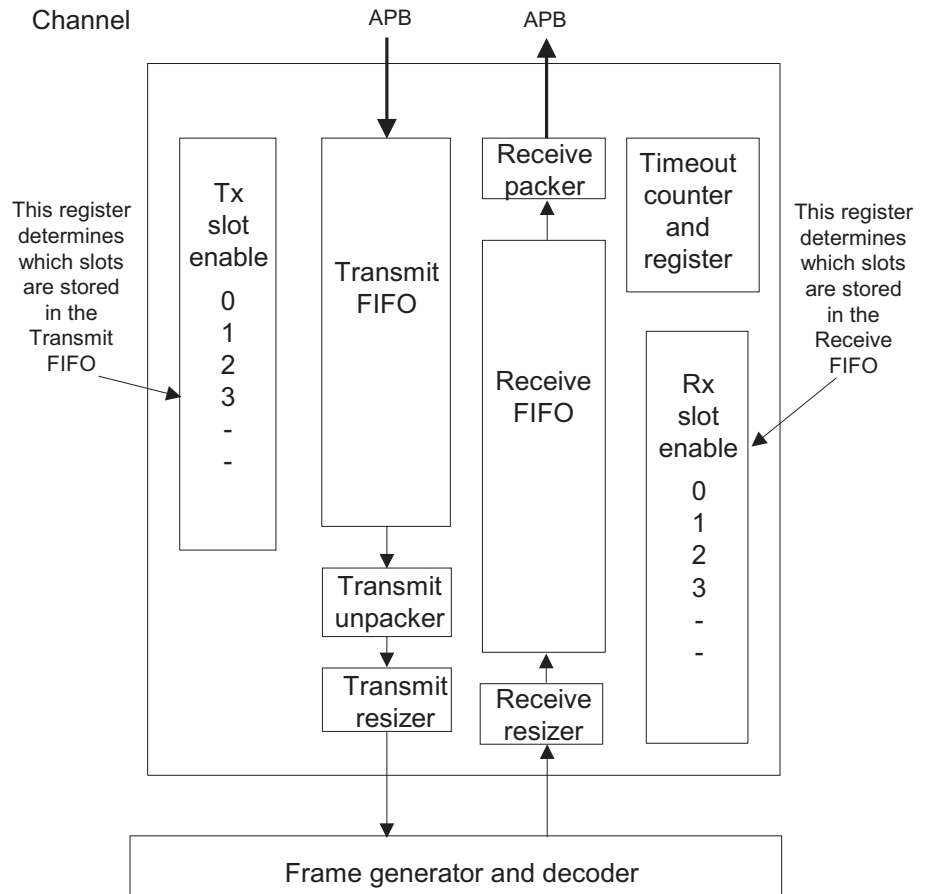


Figure 2-10 Channel block diagram

The main functional blocks of the PrimeCell AACI channel are described in the following subsections:

- *Transmit FIFOs* on page 2-18
- *Transmit unpacker* on page 2-18
- *Transmit resizer* on page 2-18
- *Receive packer* on page 2-18
- *Receive FIFOs* on page 2-19
- *Receive resizer* on page 2-19.

Transmit FIFOs

The transmit FIFOs are 20 bits wide by 8 deep. In the compact mode the FIFOs are re-organized as 40 bits wide and 4-deep. The TxCompactMode bit in the AACITXCR register determines if each 32 bit data from the CPU contains the data for two slots or one slot. If the TxCompactMode bit is set, the 32 bit data from the CPU is written into bits 31:0 of the re-organized FIFO. Data is removed from the FIFO buffers one 20-bit word at a time in non-compact mode and 16 bits at a time in compact mode. Data from the FIFO is written into the parallel-to-serial shifter.

Transmit unpacker

If the TxCompactMode bit is set to 1, a 32-bit word from the CPU to the PrimeCell AACI consists of two data slots. The PrimeCell AACI splits the 32-bit word into its two 16 bit constituents. The data in bits 15:0 are transmitted before the data in bits 31:16. You must ensure *Least Significant Bits* (LSBs) of a data word contains the lowest slot, for example, bits 15:0 contain slot 3 and bits 31:16 contain slot 4. The TxCompactMode bit only works if the TSize bits are set to either 12 or 16, so you must ensure that the AACITXCR register is programmed correctly. If the TxCompactMode bit is set to 0 then a 32-bit word into the PrimeCell AACI only contains one data slot even if TSize is set to 12 or 16. If the TxCompactMode bit is set you must ensure that the FIFO channels are set up to accept even amounts of slots, for example, slot 3 and 4, not slots 3, 4, 5, or two data elements for slot 3.

Transmit resizer

The TSize bits in the AACITXCR register determine the data size for data from the CPU. The function of the bits is to MSB justify the data received. When the data is unpacked the PrimeCell AACI left shifts the data by the desired amount to make a 20-bit word to the frame generator. The LSB bits are filled with zeros. If TSize is set to 20 no shifting or zero filling is required. The data in the FIFO is lowest slot first. For example, if the FIFO is set up to store slots 3 and 4, then slot 3 is the first data from the FIFO and slot 4 the second. If TSize is set to either 12 or 16 and operating in compact mode, then the bits 15:0 contain the lowest slot 3 and bits 31:16 slot 4.

Receive packer

The RxCompactMode bit in the AACIRXCR register determines if the 32-bit data for the CPU contains the data for one slot or two slots. If the RSize is set to 12 or 16 bits and the RxCompactMode bit is set to 1 a 32-bit word from the PrimeCell AACI consists of two data slots, the PrimeCell AACI joins two words from the FIFO into a 32-bit word. The RxCompactMode bit only works if the RSize bits are set to either 12 or 16, so you must ensure that the AACIRXCR register is programmed correctly. If the RxCompactMode bit is set to 0 then a 32-bit word from the PrimeCell AACI only

contains one data slot even if RSize is set to 12 or 16. If the RxCompactMode is set you must ensure that the FIFO channels are set up to even amounts of slots, for example slot 3 and 4, not slots 3, 4, 5, or two data elements for slot 3. In compact mode the DMA request signals are only generated when there is an even amount of data in the FIFO channel. This means that the AACIDMASREQRX in compact mode is generated when there are two words in the FIFO. You must ensure that only valid double words are read in compact mode. This is because it is possible that while a read of the FIFO is taking place, new data is received that is not valid until the second data word is received. The data from the receive channel is returned lowest slot first on read accesses from the CPU. If the receive FIFO is set up to store slots 3 and 4, then the first data word out of the FIFO is slot 3 followed by slot 4. If the RSize value is set to 12 or 16 and the RxCompactMode bit is set then slot 3 is contained in bits 15:0 and slot 4 in bits 31:16.

Receive FIFOs

The receive FIFOs are 20 bits wide by 8 deep. Bits 19:0 contain the data. You can read the AACISR register RxOverrun bit to determine if a receive overrun error occurred during reception. If the FIFOs are accessed through the APB bus it is possible that system bus latencies might not allow data to be serviced quickly enough.

Receive resizer

The RSize bits in the AACIRXCR register determine the data size for slots from the external CODEC. The RSize bits are used to LSB justify the 20 bit data from the Frame Decoder. If the RSize is set to 12, 16, or 18 bits the PrimeCell AACI must first shift right the 20-bit data from the Frame Decoder by the desired amount to form the write data for the Receive FIFO. For example, if RSize was 12, the 20-bit word from the Frame Decoder is shifted right eight places. The resizer then fills the MSB bits of the data word with zeros.

2.5 DMA interface bus protocol

The interface to the *DMA Controller* (DMAC) includes the signals listed in Table A-2 on page A-3.

The DMA signals are only connected to the receive and transmit FIFOs of channel 1 and you must ensure that the data requiring DMA is stored in these FIFOs. System requirements determine if the DMA channels are to be used.

In the receive FIFO, the DMAC monitors all of the REQ signals. When **AACIDMABREQRX** is asserted a burst of data is available and, during this time, **AACIDMASREQRX** is ignored. When the amount of data remaining is less than a burst, **AACIDMASREQRX** is asserted, allowing the DMAC to perform single transfers, as data becomes available. **AACIDMABREQRX** and **AACIDMASREQRX** need not be mutually exclusive. Essentially they are tidemarks from the FIFO, modified only to ensure that they clear for at least one cycle after a transfer has completed (as indicated by **AACIDMACLRRX**).

The PrimeCell AACI indicates to the DMAC through the **AACIDMALBREQRX** and **AACIDMALSREQRX** that the receive FIFO contains the last amount of data to be transferred before the DMAC can reassign the channel. If there has been RxTimeout in the AACISR generated, and there are only four words in the FIFO to be read by the DMAC, the **AACIDMALBREQRX** signal is asserted. If there is only one word of data in the FIFO and a RxTimeout in AACISR generated then the **AACIDMALSREQRX** is asserted. One of these two signals must be asserted in order for the DMAC to initiate the appropriate transfer then move onto the next link list item. If the FIFO is empty, and an internal Timeout (the internal Timeout uses the same TOC as RxTimeout) is generated, it is necessary for the PrimeCell AACI to tell the DMAC the last data has been sent. This is achieved through the AACIRXTOFEINTR interrupt service routine that writes to the DMAC. The **AACIDMALSREQRX** and **AACIDMALBREQRX** signals are mutually exclusive to **AACIDMASREQRX** and **AACIDMABREQRX** signals.

As the transmission data is under software control, the DMAC can be programmed with the descriptor (start location in memory, destination location, address, size of data) required for the data to be sent. When all the data is sent to the transmit FIFO and no more descriptors are set up for this channel during the transmission the DMAC can then go to the next link list item or generate a DMA interrupt.

The DMAC only transfers data to the transmit FIFO of the peripheral when the peripheral is capable of accepting a full burst of data. If the total amount of data remaining in the DMAC is less than a burst (for example, three words to be transferred in total), then the DMAC uses a series of single transfers to transfer the remaining data. The PrimeCell AACI does not need to produce **AACIDMASREQTX**.

In compact mode the DMA request signals are only generated when there is an even amount of data in the FIFO channel. This means that the **AACIDMASREQRX** in compact mode is generated when there are two words in the FIFO. You must ensure that only valid double words are read in compact mode, that is, under interrupt or DMA control. This is because it is possible that while a read of the FIFO is taking place new data can be received that is valid until the second data word is received.

The DMA interface signals are synchronous with the **PCLK** clock.

Chapter 3

Programmer's Model

This chapter describes the ARM PrimeCell AACI (PL041) registers and provides details needed when programming the microcontroller. It contains the following sections:

- *About the programmer's model* on page 3-2
- *Summary of PrimeCell AACI registers* on page 3-3
- *Register descriptions* on page 3-7
- *Interrupts* on page 3-36
- *System loopback testing* on page 3-40
- *Software restrictions* on page 3-41
- *Frequency restriction* on page 3-44.

3.1 About the programmer's model

The base address of the PrimeCell AACI is not fixed, and can be different for any particular system implementation. However, the offset of any particular register from the base address is fixed.

The following locations are reserved and must not be used during normal operation:

- locations at offsets 0x110-0x17F and 0x190-0xFDC are reserved for possible future extensions.

3.2 Summary of PrimeCell AACI registers

The PrimeCell AACI registers are shown in Table 3-1.

Table 3-1 PrimeCell AACI register summary

Address	Type	Width	Reset value	Name	Description
AACI base + 0x00	Read/write	29	0x00000000	AACIRXCR1	Control register for receive FIFO1.
AACI base + 0x04	Read/write	17	0x000000	AACITXCR1	Control register for transmit FIFO1.
AACI base + 0x08	Read	12	0x00B	AACISR1	Status register channel 1.
AACI base + 0x0C	Read	7	0x00	AACIISR1	Interrupt status channel 1.
AACI base + 0x10	Read/write	7	0x00	AACIIE1	Interrupt enable channel 1.
AACI base + 0x14	Read/write	29	0x00000000	AACIRXCR2	Control register for receive FIFO2.
AACI base + 0x18	Read/write	17	0x000000	AACITXCR2	Control register for transmit FIFO2.
AACI base + 0x1C	Read	12	0x00B	AACISR2	Status register channel 2.
AACI base + 0x20	Read	7	0x00	AACIISR2	Interrupt status channel 2.
AACI base + 0x24	Read/write	7	0x00	AACIIE2	Interrupt enable channel 2.
AACI base + 0x28	Read/write	29	0x00000000	AACIRXCR3	Control register for receive FIFO3.
AACI base + 0x2C	Read/write	17	0x000000	AACITXCR3	Control register for transmit FIFO3.
AACI base + 0x30	Read	12	0x00B	AACISR3	Status register channel 3.

Table 3-1 PrimeCell AACI register summary (continued)

Address	Type	Width	Reset value	Name	Description
AACI base + 0x34	Read	7	0x00	AACIISR3	Interrupt status channel 3.
AACI base + 0x38	Read/write	7	0x00	AACIE3	Interrupt enable channel 3.
AACI base + 0x3C	Read/write	29	0x00000000	AACIRXCR4	Control register for receive FIFO4.
AACI base + 0x40	Read/write	17	0x00000	AACITXCR4	Control register for transmit FIFO4
AACI base + 0x44	Read	12	0x00B	AACISR4	Status register channel 4.
AACI base + 0x48	Read	7	0x00	AACIISR4	Interrupt status channel 4.
AACI base + 0x4C	Read/write	7	0x00	AACIE4	Interrupt enable channel 4.
AACI base + 0x50	Read	20	0x00000	AACISL1RX	Data received on slot 1.
AACI base + 0x54	Read/write	20	0x00000	AACISL1TX	Data transmitted on slot 1.
AACI base + 0x58	Read	20	0x00000	AACISL2RX	Data received on slot 2.
AACI base + 0x5C	Read/write	20	0x00000	AACISL2TX	Data transmitted on slot 2.
AACI base + 0x60	Read	20	0x00000	AACISL12RX	Data received on slot 12.
AACI base + 0x64	Read/write	20	0x00000	AACISL12TX	Data transmitted on slot 12.
AACI base + 0x68	Read/write	14	0x0A80	AACISLFR	Slot flag register
AACI base + 0x6C	Read	8	0x00	AACISLISTAT	Slot interrupt status register.

Table 3-1 PrimeCell AACI register summary (continued)

Address	Type	Width	Reset value	Name	Description
AACI base + 0x70	Read/write	9	0x000	AACISLIEN	Slot interrupt enable register.
AACI base + 0x74	Write	13	-	AACIINTCLR	Interrupt clear register.
AACI base + 0x78	Read/write	12	0x000	AACIMAINCR	Main control register.
AACI base + 0x7C	Read/write	1	0x1	AACIRESET	RESET control register.
AACI base + 0x80	Read/write	1	0x0	AACISYNC	SYNC control register.
AACI base + 0x84	Read	28	0x00000000	AACIALLINTS	All FIFO interrupt status register.
AACI base + 0x88	Read	2	0x0	AACIMAINFR	Main flag register.
AACI base + 0x90-0xAC	Read/write	32	0x00000000	AACIDR1	Data read or written, from or to FIFO1.
AACI base + 0xB0-0xCC	Read/write	32	0x00000000	AACIDR2	Data read or written, from or to FIFO2.
AACI base + 0xD0-0xEC	Read/write	32	0x00000000	AACIDR3	Data read or written, from or to FIFO3.
AACI base + 0xF0-0x10C	Read/write	32	0x00000000	AACIDR4	Data read or written, from or to FIFO4.
AACI base + 0x110-0x17F	-	-	-	-	Reserved.
AACI base + 0x180-0x18C	-	-	-	-	Reserved for test purposes.
AACI base + 0x190-0xFDC	-	-	-	-	Reserved.
AACI base + 0xFE0	Read	8	0x41	AACIPERIPHID0	Identification register, bits 7:0.

Table 3-1 PrimeCell AACI register summary (continued)

Address	Type	Width	Reset value	Name	Description
AACI base + 0xFE4	Read	8	0x10	AACIPERIPHID1	Identification register, bits 15:8.
AACI base + 0xFE8	Read	8	0x04	AACIPERIPHID2	Identification register, bits 23:16.
AACI base + 0xFEC	Read	8	0x00	AACIPERIPHID3	Identification register bits, 31:24.
AACI base + 0xFF0	Read	8	0x0D	AACIPCELLID0	PrimeCell identification register, bits 7:0.
AACI base + 0xFF4	Read	8	0xF0	AACIPCELLID1	PrimeCell identification register, bits 15:8.
AACI base + 0xFF8	Read	8	0x05	AACIPCELLID2	PrimeCell identification register, bits 23:16.
AACI base + 0xFFC	Read	8	0xB1	AACIPCELLID3	PrimeCell identification register, bits 31:24.

Data to or from the FIFOs can be read, or written to eight sequential addresses. This allows the microprocessor to use its load and store multiple operands to read or write to the FIFO.

3.3 Register descriptions

The following registers are described in this section:

- *Data registers, AACIDR 1- 4 on page 3-8*
- *Receive control registers, AACIRXCR 1-4 on page 3-8*
- *Transmit control registers, AACITXCR 1-4 on page 3-11*
- *Status registers, AACISR 1-4 on page 3-13*
- *Interrupt status and clear registers, AACIISR 1-4 on page 3-15*
- *Interrupt enable registers, AACIIE 1-4 on page 3-16*
- *Data received on slot 1 register, AACISL1RX on page 3-17*
- *Data transmitted on slot 1 register, AACISL1TX on page 3-18*
- *Data received on slot 2 register, AACISL2RX on page 3-18*
- *Data transmitted on slot 2 register, AACISL2TX on page 3-18*
- *Data received on slot 12 register, AACISL12RX on page 3-19*
- *Data transmitted on slot 12 register, AACISL12TX on page 3-19*
- *Slot flag register, AACISLFR on page 3-20*
- *Slot interrupt status register, AACISLISTAT on page 3-22*
- *Slot interrupt enable register, AACISLIEN on page 3-23*
- *Interrupt clear register, AACIINTCLR on page 3-25*
- *Main control register, AACIMAINCR on page 3-26*
- *Reset control register, AACIRESET on page 3-29*
- *Sync control register, AACISYNC on page 3-29*
- *All FIFO interrupt status register, AACIALLINTS on page 3-30*
- *Main flag register, AACIMAINFR on page 3-30*
- *Peripheral identification registers, AACIPERIPHID0-3 on page 3-31*
- *PrimeCell identification registers, AACIPCELLID0-3 on page 3-33.*

3.3.1 Data registers, AACIDR 1- 4

The four AACIDR registers are read/write registers and are zero at reset. Table 3-2 shows the bit assignment of the AACIDR registers.

Table 3-2 AACIDR registers

Bits	Name	Type	Function
31:20	-	-	Reserved, read undefined, must be written as zeros.
19:0 (not in compact mode) 31:0 (in compact mode)	DATA	Write	Transmit FIFO. The AACITXCR register qualifies the data in the TX FIFO.
19:0 (not in compact mode) 31:0 (in compact mode)	DATA	Read	Receive FIFO. The AACIRXCR register qualifies the data in the RX FIFO.

For words to be transmitted:

- if the FIFOs are enabled, data written to this location is pushed onto the transmit FIFO
- if the FIFOs are not enabled, data is stored in the transmitter holding register (the bottom word of the transmit FIFO).

For received words:

- if the FIFOs are enabled, the data received is pushed onto the receive FIFO
- if the FIFOs are not enabled, the data received is stored in the receiving holding register (the bottom word of the receive FIFO).

The receive FIFO is 20 bits wide, the receive overrun error status can be read through the AACIISR register.

3.3.2 Receive control registers, AACIRXCR 1-4

The AACIRXCR registers are 32-bit read/write registers. The data contained in the registers controls the data slots that are contained in the receive FIFO. If more than one of the RX12-1 bits are set, then the data samples are stored sequentially in the FIFO so that the slots are received, lowest slot number first.

If data is used that is less than 20 bits wide, as set by the RSize bits, the data read from the external CODEC is automatically shifted right and the higher order bits are zero-filled. The higher order bits are up to bit 20 when not in compact mode and up to bit 16 when in compact mode. For example, if RSize is set to 18 bits then the 20-bit word from the external CODEC is shifted right by two and bits 19 and 20 filled with zeros. The data in the FIFO is stored as 20 bits and the shift and fill operations occur once the data is read from the FIFO. When you are in compact mode you must ensure that the RSize bits are set to 12 or 16. If the Compact Mode bit is set and the RSize bits are less than or equal to 16 bits, two words are read from the FIFO. If the two elements of data must be made into 16-bit words the shift and fill operations occur to give two 16-bit words. The two 16-bit words are then joined to make one 32-bit word. Table 3-3 shows the bit assignment of the AACIRXCR registers.

Table 3-3 AACIRXCR registers 1-4

Bits	Name	Type	Function
31:29	-	-	Reserved, read undefined, must be written as zeros.
28:17	TOC	Read/write	Timeout count value. The FIFOs can generate a receive timeout interrupt when the receive FIFO is not empty and no further data is received for a period of time, that is specified by the value written here. The value is the number of frames that must occur without any data being received (a count of the AACISYNC signal). A write of zero to this value disables the counter, and no time out interrupt is generated. On reset the value is zero.
16	RxFen	Read/write	FIFO enable. If this bit is set the FIFO buffers are enabled (FIFO mode). When cleared to 0 the FIFO is disabled (character mode), that is the FIFO becomes a 1-byte-deep holding register.

Table 3-3 AACIRXCR registers 1-4 (continued)

Bits	Name	Type	Function
15	RxCompact Mode	Read/write	Compact mode enable. If this bit is set to 1 and the RSize value is either 12 or 16, the two data words are compacted into a 32-bit word for reading by the CPU. If set to 0, a 32-bit word contains one slot data. In compact mode the DMA request signals are only generated when there is an even amount of data in the FIFO channel. This means that the AACIDMASREQRX in compact mode is generated when there are two words in the FIFO. You must ensure that only valid double words are read in compact mode. This is because it is possible that while a read of the FIFO is taking place new data can be received which is not valid until the second data word is received.
14:13	RSize	Read/write	00 - data is 16 bits 01 - data is 18 bits 10 - data is 20 bits 11 - data is 12 bits.
12	RX12	Read/write	FIFO stores SLOT12 data.
11	RX11	Read/write	FIFO stores SLOT11 data.
10	RX10	Read/write	FIFO stores SLOT10 data.
9	RX9	Read/write	FIFO stores SLOT9 data.
8	RX8	Read/write	FIFO stores SLOT8 data.
7	RX7	Read/write	FIFO stores SLOT7 data.
6	RX6	Read/write	FIFO stores SLOT6 data.
5	RX5	Read/write	FIFO stores SLOT5 data.
4	RX4	Read/write	FIFO stores SLOT4 data.
3	RX3	Read/write	FIFO stores SLOT3 data.
2	RX2	Read/write	FIFO stores SLOT2 data.
1	RX1	Read/write	FIFO stores SLOT1 data.
0	RxEn	Read/write	A 1 written to this bit enables the receive for this FIFO.

3.3.3 Transmit control registers, AACITXCR 1-4

The AACITXCR registers are read/write. The data contained in them controls the data slots that are contained in the transmit FIFO. The data in this FIFO must be of the same sampling frequency, for example all audio slot data at 44.1kHz. This register is used to create slot 0 for transmitting. If this register specifies that the data in it is for slot 1, 2 and 12 this takes precedence over the data in the AACISL1TX, AACISL2TX and AACISL12TX registers. You must program low-power mode through the AACISL1TX and AACISL2TX registers. If slot 1 and 2 data is to be sent through this FIFO it is always transmitted at 48kHz. It is advisable not to enable any other slots from the same channel unless they too are sampled at 48kHz. It is recommended that the FIFO is only used to transmit slot 1 and slot 2 while setting up the external CODEC after reset, after that the AACISL1TX and AACISL2TX registers are used and the FIFO used to transmit audio data.

The data contained in the TSize bits controls the number of zeros that are to be appended to data for a slot to make it 20 bits.

If the data is less than 20 bits wide as set by the TSize bits, the data read from the bus is automatically shifted left. To make a 20-bit word, the LOW bits are zero-filled. For example if TSize is set to 18 bits then to make the 20-bit word the data is shifted left by 2 and bits 0 and 1 filled with zeros. The data in the FIFO is stored as 20 bits and the shift and fill operations occur before the data is written into the FIFO. The TSize bits have a higher priority than the Compact Mode bit, so if the TSize bits are set to 18 or above the Compact Mode bit has no effect. If the Compact Mode bit is set and the TSize bits are equal to or less than 16 bits the 32-bit word from the bus is split into two 16-bit parts. These are then shifted and filled with zeros to make two 20-bit words.

You must ensure that no more than eight slots are stored in the FIFO. If you try to set up AACITXCR to allow more than eight slots to be in one transmit channel an underflow condition is signalled. Table 3-4 on page 3-12 shows the bit assignment of the AACITXCR registers.

Table 3-4 AACITXCR registers 1-4

Bits	Name	Type	Function
31:17	-	-	Reserved, read undefined, must be written as zeros.
16	TxFen	Read/write	FIFO enable. If this bit is set the FIFO buffers are enabled (FIFO mode). When cleared to 0 the FIFO is disabled (character mode), that is the FIFO becomes a 1-byte-deep holding register.
15	TxCompact Mode	Read/write	Compact mode enable. If this bit is set to 1 and the TSize value is either 12 or 16 the two data words are compacted into a 32-bit word for reading by the CPU. If set to 0, a 32-bit word contains one slot data.
14:13	TSize	Read/write	00 - data is 16 bits 01 - data is 18 bits 10 - data is 20 bits 11 - data is 12 bits.
12	TX12	Read/write	FIFO stores SLOT12 data (takes precedence over SLOT12TX).
11	TX11	Read/write	FIFO stores SLOT11 data.
10	TX10	Read/write	FIFO stores SLOT10 data.
9	TX9	Read/write	FIFO stores SLOT9 data.
8	TX8	Read/write	FIFO stores SLOT8 data.
7	TX7	Read/write	FIFO stores SLOT7 data.
6	TX6	Read/write	FIFO stores SLOT6 data.
5	TX5	Read/write	FIFO stores SLOT5 data.
4	TX4	Read/write	FIFO stores SLOT4 data.
3	TX3	Read/write	FIFO stores SLOT3 data.

Table 3-4 AACITXCR registers 1-4 (continued)

Bits	Name	Type	Function
2	TX2	Read/write	FIFO stores SLOT2 data.
1	TX1	Read/write	FIFO stores SLOT1 data (only use if sampling rate is 48kHz).
0	TxEn	Read/write	Writing 1 to this bit enables the transmit for this FIFO.

3.3.4 Status registers, AACISR 1-4

The PrimeCell AACI status registers are read/write registers that give information about the transmit or receive status of the channel. After reset the TxFF, RxFF, RxBusy, and TxBusy are 0, and TxFE and RxFE are 1. If a FIFO overrun error condition occurs an interrupt is also generated if enabled. Table 3-5 shows the bit assignment of the AACISR registers.

Table 3-5 AACISR registers 1-4

Bits	Name	Type	Function
31:12	-	Read	Reserved, read undefined, must be written as zeros.
11	RxTOFE	Read	Receive timeout FIFO empty. This bit is set to 1 if a receive time out FIFO empty has occurred and the FIFO is empty. The timeout happens when no data is received for a number of frames as specified by the time out counter (TOC bits in the AACIRXCR register). This bit is cleared to 0 by writing to the RxTOFEC bit in the AACIINTCLR register.
10	RxTIMEOUT	Read	Receive timeout. This bit is set to 1 if a receive time out has occurred and the FIFO is non-empty. The timeout happens when no data is received for the number of frames as specified by the timeout counter. This bit is cleared when the FIFO becomes empty through reading all the data (or by reading the holding register when the FIFO is disabled).

Table 3-5 AACISR registers 1-4 (continued)

Bits	Name	Type	Function
9	TxUnderrun	Read	Transmit underrun error, active HIGH. If set to 1, the PrimeCell AACI does not have enough data words in the channel for transmission. This condition only occurs if the AACISDATAIN slot1 bits 11:2 have requested data for the particular slots stored in the channel but they are not available. The underrun condition is only set when the TxEn bit in the AACITXCR is set HIGH. It remains set until the TxUEC is written to in the AACIINTCLR register, even if the TxEn bit is set back to 0 the bit does not change.
8	RxOverrun	Read	Receive overrun error. This bit is set to 1 if an overrun error is detected. This bit is set to 1 if data is received and the FIFO is already full. This bit is cleared by writing to the RxOEC bit in the AACIINTCLR register.
7	TxBusy	Read	Transmit busy, active HIGH. If set to 1, the PrimeCell AACI is busy transmitting data or the FIFO is non-empty. The TxBusy bit is set whether the transmission is enabled or not. This bit is cleared to 0 when the transmit FIFO and shift register are both empty.
6	RxBusy	Read	Receive busy, active HIGH. If set to 1, the PrimeCell AACI is busy receiving data or the FIFO is non-empty. This bit is cleared to 0 when the receive FIFO and shift register are both empty.
5	TxFF	Read	Transmit FIFO full flag, active HIGH. If the FIFO is disabled, this bit is set when the transmit holding register is full. If the FIFO is enabled, this bit is set when the transmit FIFO is full.
4	RxFF	Read	Receive FIFO full flag, active HIGH. If the FIFO is disabled, this bit is set when the receive holding register is full. This bit is asserted HIGH if the receive FIFO is full.
3	TxHE	Read	Transmit FIFO half-empty flag, active HIGH. If the FIFO is disabled, this bit is set when the transmit holding register is empty. This bit is asserted HIGH if the transmit FIFO is at least half-empty (it has space for four or more words).

Table 3-5 AACISR registers 1-4 (continued)

Bits	Name	Type	Function
2	RxHF	Read	Receive FIFO half-full flag, active HIGH. If the FIFO is disabled, this bit is set when the receive holding register is full. This bit is asserted HIGH if the receive FIFO is half or more full (it contains four or more words).
1	TxFE	Read	Transmit FIFO empty flag, active HIGH. If the FIFO is disabled, this bit is set when the transmit holding register is empty. This bit is asserted HIGH if the transmit FIFO is empty.
0	RxFE	Read	Receive FIFO empty flag, active HIGH. If the FIFO is disabled, this bit is set when the receive holding register is empty. This bit is asserted HIGH if the receive FIFO is empty.

3.3.5 Interrupt status and clear registers, AACISR 1-4

The AACISR registers are the interrupt status and clear registers for the FIFOs of the PrimeCell AACI. All bits are cleared to zero on reset except for TXCINTR. Table 3-6 shows the bit assignment of the AACISR registers.

Table 3-6 AACISR registers 1-4

Bits	Name	Type	Function
31:7	-	-	Reserved, read undefined, must be written as zeros.
6	RXTOFEINTR	Read	Receive timeout FIFO empty interrupt status. If this bit is set to 1, the receive timeout FIFO empty interrupt is asserted.
5	URINTR	Read	Transmit underflow interrupt status. If this bit is set to 1, the transmit FIFO underrun interrupt is asserted.
4	ORINTR	Read	Receive overrun interrupt status. If this bit is set to 1, the receive FIFO overrun interrupt is asserted.

Table 3-6 AACISR registers 1-4 (continued)

Bits	Name	Type	Function
3	RXINTR	Read	Receive interrupt status. If this bit is set to 1, the receive FIFO interrupt is asserted.
2	TXINTR	Read	Transmit interrupt status. If this bit is set to 1, the transmit FIFO interrupt is asserted.
1	RXTOINTR	Read	Receive timeout interrupt status. If this bit is set to 1, the receive timeout interrupt is asserted.
0	TXCINTR	Read	Transmit complete interrupt status. If this bit is set to 1, the transmit complete interrupt is asserted. No data is left to be sent from the channel (FIFO and parallel-to-serial shift register). This condition occurs when the TxBusy bit is LOW and the TxFE is HIGH in the AACISR. This bit is cleared if either of these two conditions is not true.
31:0	-	Write	A write to this register has no effect.

3.3.6 Interrupt enable registers, AACIE 1-4

The AACIE registers control the interrupt enables for the FIFOs in the PrimeCell AACI. All bits are cleared on reset. Table 3-7 shows the bit assignment of the AACIE registers.

Table 3-7 AACIE registers 1-4

Bits	Name	Type	Function
31:6	-	-	Reserved. Read undefined, must be written as zero.
6	RxTOIE	Read/write	Receive timeout FIFO empty interrupt enable. If this bit is set to 1, the receive timeout FIFO empty interrupt is enabled.
5	TxUIE	Read/write	Transmit underrun interrupt enable. If this bit is set to 1, the transmit FIFO underrun interrupt is enabled.
4	RxOIE	Read/write	Overrun receive interrupt enable. If this bit is set to 1, the receive FIFO overrun interrupt is enabled.

Table 3-7 AACIE registers 1-4

Bits	Name	Type	Function
3	RxIE	Read/write	Receive interrupt enable. If this bit is set to 1, the receive FIFO interrupt is enabled.
2	TxIE	Read/write	Transmit interrupt enable. If this bit is set to 1, the transmit FIFO interrupt is enabled.
1	RxTIE	Read/write	Receive timeout interrupt enable. If this bit is set to 1, the receive timeout interrupt is enabled.
0	TxCIE	Read/write	Transmit complete interrupt enable. If this bit is set to 1, the transmit complete interrupt is enabled. No data is left in the channel for transmit. (Both the FIFO and the parallel-to-serial shift register are empty).

3.3.7 Data received on slot 1 register, AACISL1RX

The AACISL1RX register is a read-only register. This register contains the last valid data sent over the AC-link from slot 1 provided the S11RxEn bit is set in the AACIMAINCR register. Table 3-8 shows the bit assignment of the AACISL1RX register.

Table 3-8 AACISL1RX register

Bits	Name	Type	Function
31:20	-	-	Reserved, read undefined, must be written as zeros.
19:0	DATA	Read	Read data value of the last value written to this register through the AC-link interface. The data contained in it, is the last valid received slot 1. For example, the slot 0 received tagged slot1 as valid because the sample rate request bits can change without slot 0 tagging the slot 1 valid. The tag bit is only set when slot 1 returns a valid address.

————— **Note** —————

Sample rate request bits are allowed to change without slot 1 being marked as valid in the receive slot 0.

3.3.8 Data transmitted on slot 1 register, AACISL1TX

The AACISL1TX register is a read/write register. When a write occurs to this register the data contained is sent on the next available frame in slot 1. If a power down is required, then data must be written to the address 0x26 on the external CODEC, which is recorded by the PrimeCell AACI. If the AACISL2TX bit 16 is set, then the PrimeCell AACI goes into power down mode. Data in the AACISL2TX is only sent when there is valid data written into AACISL1TX. You must ensure that AACISL2TX is always written to, if required, before data is written to AACISL1TX. AACISL2TX has a dependency on AACISL1TX. The last data value written to this register is the data value that is read. Table 3-9 shows the bit assignment of the AACISL1TX register.

Table 3-9 AACISL1TX register

Bits	Name	Type	Function
31:20	-	-	Reserved, read undefined, must be written as zeros.
19:0	DATA	Read/write	Write data value to transmit on slot1 of the next available frame. Once the data is transmitted it is marked as invalid. The data read from this register is the last data written to it.

3.3.9 Data received on slot 2 register, AACISL2RX

The AACISL2RX register is a read-only register. This register contains the last valid data sent over the AC-link from slot 2. Table 3-10 shows the bit assignment of the AACISL2RX register.

Table 3-10 AACISL2RX register

Bits	Name	Type	Function
31:20	-	-	Reserved, read undefined, must be written as zeros.
19:0	DATA	Read/write	Read data value of the last value written to this register through the AC-link interface.

3.3.10 Data transmitted on slot 2 register, AACISL2TX

The AACISL2TX register is a read/write register. When a write occurs to this register the data it contains is sent on the next available frame in slot 2. If a power down is required, then data must be written to AACISL1TX location address 0x26, which is

recorded by the PrimeCell AACI. If the AACISL2TX bit 16 is set, then the PrimeCell AACI goes into power down mode. Table 3-11 shows the bit assignment of the AACISL2TX register.

Table 3-11 AACISL2TX register

Bits	Name	Type	Function
31:20	-	-	Reserved, read undefined, must be written as zeros.
19:0	DATA	Read/write	Write data value to transmit on slot 2 of the next available frame. Once the data is transmitted it is marked as invalid. The data read from this register is the last data written to it.

3.3.11 Data received on slot 12 register, AACISL12RX

The AACISL12RX register is a read-only register. When this register is read the data contained in it is the data that was last received for slot 12. Table 3-12 shows the bit assignment of the AACISL12RX register.

Table 3-12 AACISL12RX register

Bits	Name	Type	Function
31:20	-	-	Reserved, read undefined, must be written as zeros.
19:0	DATA	Read	Read data value of the last valid slot 12 data received from the AC-link. Bit 0 is monitored to generate the AACIGPIOINTR interrupt. In order to monitor bit 0, you must enable the receive logic by setting the SI12RxEn bit in the AACIMAINCR register.

3.3.12 Data transmitted on slot 12 register, AACISL12TX

The AACISL12TX register is a 20-bit read/write register. When a write occurs to this register the data contained in it is sent on the next available frame in slot 12. When this register is read it contains the data that was last written to it. Table 3-13 on page 3-20 shows the bit assignment of the AACISL12TX register.

Table 3-13 AACISL12TX register

Bits	Name	Type	Function
31:20	-	-	Reserved, read undefined, must be written as zeros.
19:0	DATA	Read/write	Write data value to transmit on slot 12 of the next available frame. Once the data is transmitted it is marked as invalid. The data read from this register is the last data written to it.

3.3.13 Slot flag register, AACISLFR

The AACISLFR register is a read-only register that gives the raw status of various functions outside of the FIFO functionality in the PrimeCell AACI. The bits in this register generate interrupts if the appropriate AACISLIEN bits are set. All bits are zero at reset except for the EMPTY flags which are 1.

Table 3-14 shows the bit assignment of the SLOT2FR register.

Table 3-14 AACISLFR register

Bits	Name	Type	Function
31:14	-	Read	Reserved, read undefined, must be written as zeros.
13	RWIS	Read	RAW wake-up interrupt status. If this bit is set to 1 the synchronized wake-up interrupt is asserted. The WISE enable bit in the AACISLIEN (slot interrupt enable register) must be set to one. This bit is cleared with a write of 1 to the AACIINTCLR register. If the WISE bit is set to 0 and no PCLK is running, this register does not detect the assertion of the wake-up interrupt. In order to detect the wake-up interrupt the LowPower Mode bit in the AACIMAINCR register must be set.
12	RawGPIOINTR	Read	RAW GPIO interrupt. The RawGPIOINTR shows the raw status of the GPIOINT bit (slot 12 bit 0) in the receive frame, which is stored in the AACISL12RX register. This bit is cleared when the AACISL12RX register is read.

Table 3-14 AACISLFR register (continued)

Bits	Name	Type	Function
11	SI12TxEmpty	Read	Slot 12 transceive empty. The SI12TxEmpty bit is set when there is no data present in the AACISL12TX register.
10	SI12RxValid	Read	Slot 12 receive valid. The SI12RxValid bit is set when the AACISL12RX register is full.
9	SI2TxEmpty	Read	Slot 2 transceive empty. The SI2TxEmpty bit is set when there is no data present in the AACISL2TX register.
8	SI2RxValid	Read	Slot 2 receive valid. The SI2RxValid bit is set when the AACISL2RX register is full.
7	SI1TxEmpty	Read	Slot 1 transceive empty. The SI1TxEmpty bit is set when there is no data present in the AACISL1TX register.
6	SI1RxValid	Read	Slot 1 receive valid. The SI1RxValid bit is set when the AACISL1RX register is full.
5	SI12TxBusy	Read	Slot 12 transceive busy. If set to 1, the data in the AACISL12TX register is busy being transmitted or the register is non-empty. This bit is cleared when the data in the register is transmitted and the register is empty. The SI12TxBusy bit is set whether transmission is enabled or not.
4	SI12RxBusy	Read	Slot 12 receive busy. If set to 1, the AACISL12RX register is busy receiving new data or the register is non-empty. This bit is cleared to zero when the receiver is empty and the receive logic is not active, or receive is disabled.
3	SI2TxBusy	Read	Slot 2 transceive busy. If set to 1, the data in the AACISL2TX register is busy being transmitted or the register is non-empty. This bit is cleared when the data in the register is transmitted and the register is empty. The SI2TxBusy bit is set whether transmission is enabled or not.

Table 3-14 AACISLFR register (continued)

Bits	Name	Type	Function
2	Sl2RxBusy	Read	Slot 2 receive busy. If set to 1, the AACISL2RX register is busy receiving new data or the register is non-empty. This bit is cleared to zero when the receiver is empty and the receive logic is not active.
1	Sl1TxBusy	Read	Slot 1 transceive busy. If set to 1, the data in the AACISL1TX register is busy being transmitted or the register is non-empty. This bit is cleared when the data in the register is transmitted and the register is empty. The Sl1TxBusy bit is set whether transmission is enabled or not.
0	Sl1RxBusy	Read	Slot 1 receive busy. If set to 1, the AACISL1RX register is busy receiving new data or the register is non-empty. This bit is cleared to zero when the receiver is empty and the receive logic is not active.

3.3.14 Slot interrupt status register, AACISLISTAT

The AACISLISTAT register is the interrupt status register for the AACISL1TX, AACISL2TX, AACISL12TX, AACISL1RX, AACISL2RX, and AACISL12RX registers, and for the external wake-up interrupt. The values in this register are derived from the values in the AACISLFR, the AACISLIEN register. All bits are cleared to zero on reset. Table 3-15 shows the bit assignment of the AACISLISTAT register.

Table 3-15 AACISLISTAT register

Bits	Name	Type	Function
31:8	-	Read	Reserved, read undefined, must be written as zeros.
7	WIS	Read	Wake-up interrupt status. If this bit is set to 1 the wake-up interrupt is asserted.
6	GPIOINTR	Read	GPIO interrupt status. If this bit is set to 1 the AACIGPIOINTR interrupt is asserted.
5	S12TXINTR	Read	Slot 12 transmit interrupt status. If this bit is set to 1, AACIS12TXINTR interrupt is asserted.

Table 3-15 AACISLISTAT register (continued)

Bits	Name	Type	Function
4	S12RXINTR	Read	Slot 12 receive interrupt status. If this bit is set to 1, AACIS12RXINTR interrupt is asserted.
3	S2TXINTR	Read	Slot 2 transmit interrupt status. If this bit is set to 1, AACIS2TXINTR interrupt is asserted.
2	S2RXINTR	Read	Slot 2 receive interrupt status. If this bit is set to 1, AACIS2RXINTR interrupt is asserted.
1	S1TXINTR	Read	Slot 1 transmit interrupt status. If this bit is set to 1, AACIS1TXINTR interrupt is asserted.
0	S1RXINTR	Read	Slot 1 receive interrupt status. If this bit is set to 1, AACIS1RXINTR interrupt is asserted.

3.3.15 Slot interrupt enable register, AACISLIEN

The AACISLIEN register is a read/write register that controls the interrupt enables for the interrupts outside the FIFO channels. All bits are cleared to zero on reset. Table 3-16 shows the bit assignment of the AACISLIEN register.

Table 3-16 AACISLIEN register

Bits	Name	Type	Function
31:9	-	Read/write	Reserved. read undefined must be written as zeros.
8	WISE	Read/write	Wake-up synchronization enable. If this bit is set to 1 the wake-up interrupt is synchronized to PCLK , (PCLK must be running to allow interrupt to be detected in this mode.) If this bit is set to 0 the wake-up interrupt is not synchronized to PCLK and is fed straight to the interrupt controller. You must ensure that the interrupt controller logic synchronizes the wake-up interrupt in this mode. This mode allows the wake-up interrupt to be detected without PCLK running which saves the power consumption.
7	WakeupIE	Read/write	Wake-up interrupt enable. If this bit is set to 1 the wake-up interrupt is enabled.
6	GpioIE	Read/write	GPIO interrupt enable. If this bit is set to 1, the AACIGPIOINTR interrupt is enabled.
5	Slot12TxIE	Read/write	Slot 12 transmit interrupt enable. If this bit is set to 1, AACIS12TXINTR interrupt is enabled.
4	Slot12RxIE	Read/write	Slot 12 receive interrupt enable. If this bit is set to 1, AACIS12RXINTR interrupt is enabled.
3	Slot2TxIE	Read/write	Slot 2 transmit interrupt enable. If this bit is set to 1, AACIS2TXINTR interrupt is enabled.
2	Slot2RxIE	Read/write	Slot 2 receive interrupt enable. If this bit is set to 1, AACIS2RXINTR interrupt is enabled.
1	Slot1TxIE	Read/write	Slot 1 transmit interrupt enable. If this bit is set to 1, AACIS1TXINTR interrupt is enabled.
0	Slot1RxIE	Read/write	Slot 1 receive interrupt enable. If this bit is set to 1, AACIS1RXINTR interrupt is enabled.

3.3.16 Interrupt clear register, AACIINTCLR

The AACIINTCLR register is a write-only register that allows the WIS, TXUE, RXTOFE, and RXOE interrupts to be cleared. Table 3-17 shows the bit assignment of the AACIINTCLR register.

Table 3-17 AACIINTCLR register

Bits	Name	Type	Function
31:13	-	-	Reserved, read undefined, must be written as zeros.
12	RxTOFEC4	Write	Receive timeout FIFO empty clear. A write of one to this bit clears the RxTOFEC4 bit in the AACISR register.
11	RxTOFEC3	Write	Receive timeout FIFO empty clear. A write of one to this bit clears the RxTOFEC3 bit in the AACISR register.
10	RxTOFEC2	Write	Receive timeout FIFO empty clear. A write of one to this bit clears the RxTOFEC2 bit in the AACISR register.
9	RxTOFEC1	Write	Receive timeout FIFO empty clear. A write of one to this bit clears the respective RxTOFEC1 bit in the AACISR register.
8	TxUEC4	Write	Transmit underrun error clear. A write of one to this bit clears the TxUEC4 bit in the AACISR register.
7	TxUEC3	Write	Transmit underrun error clear. A write of one to this bit clears the TxUEC3 bit in the AACISR register.
6	TxUEC2	Write	Transmit underrun error clear. A write of one to this bit clears the TxUEC2 bit in the AACISR register.
5	TxUEC1	Write	Transmit underrun error clear. A write of one to this bit clears the TxUEC1 bit in the AACISR register.
4	RxOEC4	Write	Receive overrun error clear. A write of one to this bit clears the RxOEC4 bit in the AACISR register.
3	RxOEC3	Write	Receive overrun error clear. A write of one to this bit clears the RxOEC3 bit in the AACISR register.

Table 3-17 AACIINTCLR register (continued)

Bits	Name	Type	Function
2	RxOEC2	Write	Receive overrun error clear. A write of one to this bit clears the RxOEC2 bit in the AACISR register.
1	RxOEC1	Write	Receive overrun error clear. A write of one to this bit clears the RxOEC1 bit in the AACISR register.
0	WISC	Write	Wake-up interrupt status clear. A write of 1 to this bit clears the WIS interrupt bit.

3.3.17 Main control register, AACIMAINCR

The AACIMAINCR register is the main control register for the PrimeCell AACI. All bits are cleared on reset. Table 3-18 shows the bit assignment of the AACIMAINCR register.

Table 3-18 AACIMAINCR register

Bits	Name	Type	Function
31:12	-	-	Reserved, read undefined, must be written as zeros.
11:10	SCRA[1:0]	Read/write	Secondary CODEC register access.
9	DMAEnable	Read/write	DMA enable. If this bit is set to 1, the DMA logic is enabled. Defaults to 0 on reset.
8	Sl12TxEn	Read/write	Slot 12 transmit enable. If this bit is set to 1 the SLOT12TX transmit logic is enabled. Defaults to 0 on reset. When set to 0 the transmit logic is disabled for the slot register. You must ensure that the Busy bit for this register is not active before disabling the logic.
7	Sl12RxEn	Read/write	Slot 12 receive enable. If this bit is set to 1 the AACISL12RX receive logic is enabled. Defaults to 0 on reset. When set to 0 the receive logic is disabled for the slot register. You must ensure that the Busy bit for this register is not active before disabling the logic.

Table 3-18 AACIMAINCR register (continued)

Bits	Name	Type	Function
6	Sl2TxEn	Read/write	Slot 2 transmit enable. If this bit is set to 1 the AACISL2TX transmit logic is enabled. Defaults to 0 on reset. When set to 0 the transmit logic is disabled for the slot register. You must ensure that the Busy bit for this register is not active before disabling the logic.
5	Sl2RxEn	Read/write	Slot 2 receive enable. If this bit is set to 1 the AACISL2RX receive logic is enabled. Defaults to 0 on reset. When set to 0 the receive logic is disabled for the slot register. You must ensure that the Busy bit for this register is not active before disabling the logic.
4	Sl1TxEn	Read/write	Slot 1 transmit enable. If this bit is set to 1 the AACISL1TX transmit logic is enabled. Defaults to 0 on reset. When set to 0 the transmit logic is disabled for the slot register. You must ensure that the Busy bit for this register is not active before disabling the logic.
3	Sl1RxEn	Read/write	Slot 1 receive enable. If this bit is set to 1, the AACISL1RX receive logic is enabled. Defaults to 0 on reset. When set to 0 the receive logic is disabled for the slot register. You must ensure that the Busy bit for this register is not active before disabling the logic.

Table 3-18 AACIMAINCR register (continued)

Bits	Name	Type	Function
2	LowPower Mode	Read/write	Low-power mode. You must set this bit HIGH once the CODEC is programmed to enter the PrimeCell AACI into low-power mode. (Power down register 0x26 is programmable to the appropriate value through the AACISL1TX/AACISL2TX registers.) You must ensure that this bit is only set after the CODEC is programmed into low-power mode, plus the time it takes the AACIBITCLK and AACISDATAIN signals to go LOW (normally the TS2_PDOWN parameter in CODEC datasheets). You must ensure that the receive logic and the FIFO transmit logic is disabled and not BUSY before low-power mode is entered. Once this bit is set you can monitor wake-up events on the AACISDATAIN lines. When the CODEC is woken up this bit must be set LOW.
1	LoopBack	Read/write	Loopback mode. If this is set to 1, loopback test mode is enabled. Defaults to 0 when reset by PRESETn . You must ensure this bit is always 0 for normal operation. When LoopBack is set, the CODEC can be assumed to be ready.
0	AacIfE	Read/write	PrimeCell AACI interface enable. If this bit is set, the PrimeCell AACI is enabled. Defaults to 0 on reset. When set to 0 the PrimeCell AACI is held in its initialized state, however, the registers can be accessed as normal.

3.3.18 Reset control register, AACIRESET

The AACIRESET register is a read/write register that controls the **AACIRESET** (output signal to the CODEC). The register bit is set to 1 when reset. The **AACIRESET** port is set HIGH when the PrimeCell AACI is reset. Table 3-19 shows the bit assignment of the AACIRESET register.

Table 3-19 AACIRESET register

Bits	Name	Type	Function
31:1	-	-	Reserved, read undefined, must be written as zeros.
0	Forced RESET	Read/write	Forced reset. The AACIRESET port follows whatever value is written to this bit. You must ensure that the signal is LOW long enough to meet the specification of the external device. The AACIRESET port is active LOW.

3.3.19 Sync control register, AACISYNC

The AACISYNC register is a read/write register that controls the **AACISYNC** (output signal to the CODEC). The register bit is cleared to 0 when reset. The **AACISYNC** port is set LOW when the PrimeCell AACI is reset. Table 3-20 shows the bit assignment of the AACISYNC register.

Table 3-20 AACISYNC register

Bits	Name	Type	Function
31:1	-	-	Reserved, read undefined, must be written as zeros.
0	Forced SYNC	Read/write	Forced sync. The normal AACISYNC output can be overridden using this bit. You must ensure that the signal is HIGH long enough to meet the specification of the external device. You must only use this register when the external CODEC is in low-power mode. The AACISYNC port is active HIGH.

3.3.20 All FIFO interrupt status register, AACIALINTS

The AACIALINTS register is read-only, and echoes all the FIFO interrupt status registers in the PrimeCell AACI. This register allows all the FIFO interrupt sources to be read with one read. Table 3-21 shows the bit assignment of the AACIALINTS register.

Table 3-21 AACIALINTS register

Bits	Name	Type	Function
31:28	-	-	Reserved, read undefined, must be written as zeros.
27:21	AACIISR4	Read	Copy of the AACIISR4 register.
20:14	AACIISR3	Read	Copy of the AACIISR3 register.
13:7	AACIISR2	Read	Copy of the AACIISR2 register.
6:0	AACIISR1	Read	Copy of the AACIISR1 register.

3.3.21 Main flag register, AACIMAINFR

The AACIMAINFR register is a read-only register. This register is 0 at reset. Table 3-22 shows the bit assignment of the AACIMAINFR register.

Table 3-22 AACIMAINFR register

Bits	Name	Type	Function
31:2	-	-	Reserved, read undefined, must be written as zeros.
1	MainTxBusy	Read	Main transmit busy. If this bit is set to 1, the PrimeCell AACI transmit logic is busy. This bit is cleared when all the TxBusy bits for slot registers and FIFOs are 0.
0	MainRxBusy	Read	Main receive busy. If this bit is set to 1 the PrimeCell AACI receive logic is busy. This bit is cleared when all the RxBusy bits for slot registers and FIFOs are 0.

When all the channels and the slot registers are no longer busy the MainTxBusy/MainRxBusy bits return a LOW.

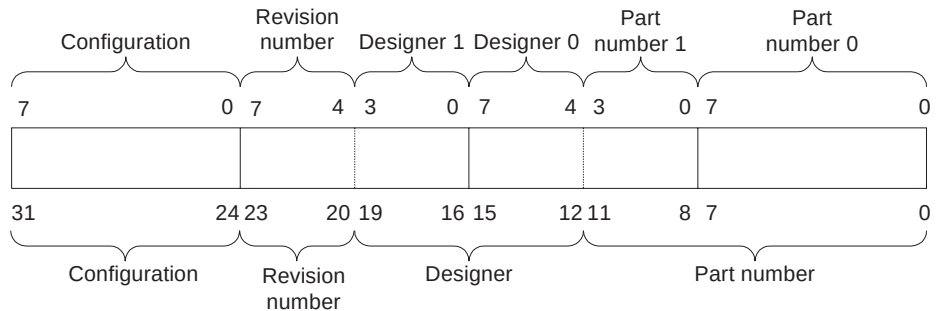
3.3.22 Peripheral identification registers, AACIPERIPHID0-3

The peripheral identification registers are four 8-bit registers. The registers can conceptually be treated as a single 32-bit register. The read-only registers provide the following options of the peripheral:

- Part number[11:0]** This is used to identify the peripheral. The three digits Product Code 0x041 is used.
- Designer[19:12]** This is the identification of the designer. ARM Ltd. is 0x41 (ASCII A).
- Revision[23:20]** This is the revision number of the peripheral. The revision number starts from 0.
- Configuration[31:24]** This the configuration options of the peripheral. The configuration value is 0.

Figure 3-1 shows the bit allocation for the peripheral identification register.

Actual register bit assignment



Conceptual register bit assignment

Figure 3-1 Peripheral identification register bit allocation

Note

When you design a systems memory map you must remember that the register has a 4KB-memory footprint.

The 4-bit revision number is implemented by instantiating a component called *RevisionAnd* four times with its inputs tied off as appropriate, and the output sent to the

read multiplexor.

All memory accesses to the peripheral identification registers must be 32-bit, using the LDR and STR instructions.

The four, 8-bit peripheral identification registers are described in the following subsections:

- *AACIPERIPHID0 register*
- *AACIPERIPHID1 register*
- *AACIPERIPHID2 register* on page 3-33
- *AACIPERIPHID3 register* on page 3-33.

AACIPERIPHID0 register

The peripheral identification register AACIPERIPHID0 is hard coded and the fields within the register determine the reset value. Table 3-23 shows the bit assignment of the AACIPERIPHID0 register.

Table 3-23 AACIPERIPHID0 register

Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:0	Partnumber0	Read	These bits read back as 0x41.

AACIPERIPHID1 register

The peripheral identification register AACIPERIPHID1 is hard coded and the fields within the register determine the reset value. Table 3-24 shows the bit assignment of the AACIPERIPHID1 register.

Table 3-24 AACIPERIPHID1 register

Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:4	Designer0	Read	These bits read back as 0x1.
3:0	Partnumber1	Read	These bits read back as 0x0.

AACIPERIPHID2 register

The peripheral identification register AACIPERIPHID2 is hard coded and the fields within the register determine the reset value. Table 3-25 shows the bit assignment of the AACIPERIPHID0 register.

Table 3-25 AACIPERIPHID2 register

Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:4	Revision	Read	These bits read back as 0x0.
3:0	Designer1	Read	These bits read back as 0x4.

AACIPERIPHID3 register

The peripheral identification register AACIPERIPHID3 is hard coded and the fields within the register determine the reset value. Table 3-26 shows the bit assignment of the AACIPERIPHID3 register.

Table 3-26 AACIPERIPHID3 register

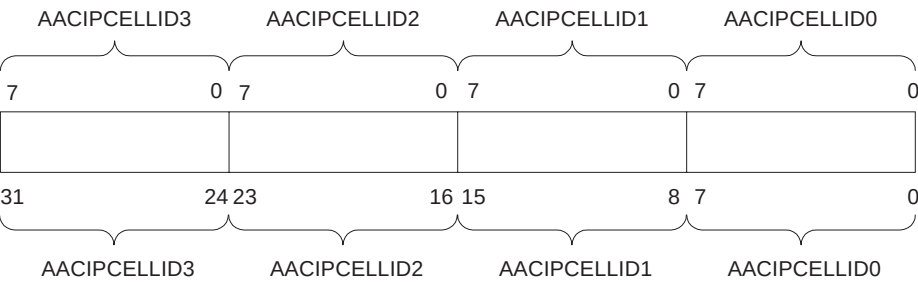
Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:0	Configuration	Read	These bits read back as 0x00.

3.3.23 PrimeCell identification registers, AACIPCELLID0-3

The PrimeCell identification registers are four 8-bit registers. The read-only registers can conceptually be treated as a single 32-bit register. The register is used by the PrimeCell BIOS for automatic peripheral configuration. AACIPCELLID, this register is set to 0xB105F00D.

Figure 3-2 shows the bit allocation for the PrimeCell identification registers.

Actual register bit assignment



Conceptual register bit assignment

Figure 3-2 PrimeCell identification register bit allocation

The four, 8-bit registers are described in the following subsections:

- *AACIPCELLID0 register*
- *AACIPCELLID1 register* on page 3-35
- *AACIPCELLID2 register* on page 3-35
- *AACIPCELLID3 register* on page 3-35.

AACIPCELLID0 register

The PrimeCell identification register AACIPCELLID0 is hard coded and the fields within the register determine the reset value. Table 3-27 shows the bit assignment of the AACIPCELLID0 register.

Table 3-27 AACIPCELLID0 register

Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:0	PrimeCellID0	Read	These bits read back as 0x0D.

AACIPCELLID1 register

The PrimeCell identification register AACIPCELLID1 is hard coded and the fields within the register determine the reset value. Table 3-28 shows the bit assignment of the PrimeCellID1 register.

Table 3-28 AACIPCELLID1 register

Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:0	PrimeCellID1	Read	These bits read back as 0xF0.

AACIPCELLID2 register

The PrimeCell identification register AACIPCELLID2 is hard coded and the fields within the register determine the reset value. Table 3-29 shows the bit assignment of the AACIPCELLID2 register.

Table 3-29 AACIPCELLID2 register

Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:0	PrimeCellID2	Read	These bits read back as 0x05.

AACIPCELLID3 register

The PrimeCell identification register AACIPCELLID3 is hard coded and the fields within the register determine the reset value. Table 3-30 shows the bit assignment of the AACIPCELLID3 register.

Table 3-30 AACIPCELLID3 register

Bits	Name	Type	Function
31:8	-	-	Reserved, read undefined, must be written as zeros.
7:0	PrimeCellID3	Read	These bits read back as 0xB1.

3.4 Interrupts

The PrimeCell AACI generates individual maskable active HIGH interrupts. You can enable and disable each interrupt by changing the appropriate enable bit. Setting the bit HIGH enables the corresponding interrupt. This allows for a system interrupt controller to provide the mask registers for each interrupt.

The interrupts are also output as a combined single interrupt that is an OR function of the individual masked sources. AACIINTR is asserted if any of the individual interrupts are asserted and unmasked. You can connect this output to the system interrupt controller to provide another level of masking on a per-peripheral basis. This allows you to use modular device drivers that always know where to find the interrupt source control register bits. The status of the individual interrupt sources can be read from the appropriate register.

3.4.1 Interrupt generation logic

The maskable active HIGH interrupts and the combined single interrupt AACIINTR that are generated by the PrimeCell AACI are described below:

AACIRXINTR 1-4 If the FIFO is enabled, the FIFO receive interrupt is asserted when the PrimeCell AACI receive FIFO is greater than, or equal to, half full and the mask bit RxIE is set. The receive interrupt is cleared when the FIFO becomes less than half full.

If the FIFOs are disabled (have a depth of one location) and data is received, thereby filling the location, the receive interrupt is asserted HIGH. In this case the receive interrupt is cleared by performing a single read of the receive FIFO.

AACITXINTR 1-4 If the FIFO is enabled, the FIFO transmit interrupt is asserted when the PrimeCell AACI transmit FIFO is less than, or equal to, half-empty and the mask bit TxIE is set. The FIFO transmit interrupt is cleared by filling the transmit FIFO to more than half full.

If the FIFOs are disabled (have a depth of one location) and there is no data present in the transmitters single location, the transmit interrupt is asserted HIGH. In this case the transmit interrupt is cleared by performing a single write to the transmit FIFO.

AACIORINTR 1-4 If a receive FIFO has an overrun condition, the AACIORINTR is asserted HIGH. Performing a write of 1 to the RxOEC bit in the AACIINTCLR register clears the interrupt. The receiver overrun condition is set when the receive logic attempts to place data into the receive buffer after it is filled.

When overrun occurs, the contents of the FIFO are not cleared.

Only the last received word that triggered the overrun condition is lost. You must read out the data in the RX FIFO to clear the FIFO. Alternatively, if the PrimeCell AACI interface enable bit in the AACIMAINCR register is cleared, all the FIFOs are cleared. If the FIFOs are disabled (have a depth of one location), and the receive logic attempts to place data into this buffer after it is filled, the interrupt is generated.

AACIURINTR 1-4 If a transmit FIFO has an underrun condition, the AACIURINTR is asserted HIGH. Performing a write of 1 to the TxUEC bit in the AACIINTCLR register clears the interrupt. The transmit underrun condition is set when the external CODEC has requested data but the channel the data is transmitted from does not contain the correct number of data words to be transmitted. For example, the channel is programmed, through the AACITXCR register to contain data for slot 2 and slot 3 only, the CODEC requests data for both these slots (as both must be at the same sample rate) but the FIFO in the channel only has slot 2 data. If this condition occurred a transmit underrun error would occur. The transmit underrun condition is qualified by the TxEn bit in the AACITXCR. When TxEn is set to 0 the underrun condition is not set. However the TxEn going to 0 does not clear the AACIURINTR.

AACIRXTOINTR 1-4

The receive timeout interrupt is asserted when the receive FIFO is not empty and no further data is received over a number of frames set by the TOC value in the AACIRXCR register. The receive timeout interrupt is cleared when the FIFO becomes empty through reading all the data.

AACIRXTOFEINTR 1-4

The receive timeout FIFO empty interrupt is asserted when the receive FIFO is empty and no further data is received over a number of frames set by the TOC value in the AACIRXCR register. The receive timeout FIFO empty interrupt is cleared by writing 1 to the AACIINTCLR register bit RxTOFEC. If one of the DMA signals, or the AACIRXTOINTR is raised, it is not followed by the receive timeout FIFO empty interrupt.

AACITXCINTR 1-4

The transmit complete interrupt is asserted when the transmit FIFO is empty and the parallel-to-serial shifter is empty. This indicates that there is no data left in the FIFOs to be sent. This

condition occurs when the TxBusy bit is LOW and the TxFE is HIGH in the AACISR. This bit is cleared if either of these two conditions is not true.

- AACIWINTR** The wake-up interrupt is asserted when a wake-up event triggers the assertion of **AACISDATAIN** while the AC-link is powered down. The wake-up is caused by the GPIO pins of the external CODECs, which have been configured to generate a wake-up event through the GPIO PIN WAKE-UP CONTROL register (52H). An AC-link wake-up interrupt is defined as a 0 to 1 transition on **AACISDATAIN** when the AC-link is powered down. When the wake event is detected on the **AACISDATAIN** line an interrupt is generated to allow the processor to reactivate the link with either a warm or cold reset.
- AACIGPIOINTR** The receive AACIGPIOINTR interrupt is asserted when bit 0 in slot 12 of the incoming **AACISDATAIN** is 1. In order to monitor bit 0, you must enable the receive logic, by setting the S12RxEn bit in the AACIMAINCR register. This bit indicates that one or more of the bits in slot 12 have changed since the last frame. The interrupt service routine must read the AACISL12RX register in order to clear this interrupt. Register 54H, GPIO PIN STATUS, of the external CODECs reflects the state of all the GPIO pins.
- AACIS12RXINTR** The receive AACIS12RXINTR interrupt is asserted when the AACISL12RX register has new data that has not been read. By reading the data in the AACISL12RX register the AACIS12RXINTR interrupt is cleared.
- AACIS12TXINTR** The transmit AACIS12TXINTR interrupt is asserted HIGH when there is no data present in the AACISL12TX register. It is cleared by performing a single write to the AACISL12TX register.
- AACISRXINTR** The receive AACIS2RXINTR interrupt is asserted when the AACISL2RX register has new data that has not been read. By reading the data in the AACISL2RX register the AACISL2RXINT interrupt is cleared.
- AACIS2TXINTR** The transmit AACIS2TXINTR interrupt is asserted HIGH when there is no data present in the AACISL2TX register. It is cleared by performing a single write to the AACISL2TX register.
- AACIS1RXINTR** The receive AACIS1RXINTR interrupt is asserted when the AACISL1RX register has new data that has not been read. By reading the data in the AACISL1RX register the AACIS1RXINTR interrupt is cleared.

AACIS1TXINTR The transmit AACIS1TXINTR interrupt is asserted HIGH when there is no data present in the AACISL1TX register. It is cleared by performing a single write to the AACISL1TX register.

The individual masked interrupt outputs are also combined into a single output that is an OR function of the individual sources. This output can be connected to the system interrupt controller to provide another level of masking on an individual per-peripheral basis.

AACIINTR The combined PrimeCell AACI interrupt is asserted when any of the individual interrupts above are asserted and the corresponding mask is enabled.

The interrupt AACITXINTR is not qualified with the TxEN bit in the AACITXCR register, which allows operation in one of two ways. Data can be written to the transmit FIFOs prior to enabling the PrimeCell AACI, and then interrupts can be enabled. Alternatively, the PrimeCell AACI can be transmit enabled and interrupt enabled so that data values are written to the transmit FIFO by the interrupt service routine or the DMA controller. The FIFOs are flushed when the AacIfE bit in the AACIMAINCR is deasserted.

3.5 System loopback testing

A loopback test mode is available for system testing so that data transmitted on **AACISDATAOUT** can also be received on **AACISDATAIN**. Loopback mode is entered when a 1 is written to the LoopBack bit in AACIMAINCR. For normal operation the LoopBack bit must always be 0, that is also the default state at reset. When LoopBack is set, the CODEC can be assumed to be ready. You must ensure that the receive logic is enabled before the transmission logic to ensure that no data is lost.

3.6 Software restrictions

You must ensure:

- The AACI channel control registers are not written into during transmission, or reception. Software must poll for the TxBusy and RxBusy bits in the AACISR1-4 to be cleared before writing into the AACIRXCR1-4 and AACITXCR1-4 registers.
- The AACITXCR1-4 and AACIRXCR1-4 registers are not programmed for more than eight slots per channel at a time.
- Once a value has been written into a AACITXCR1-4 register, the AACI ignores further writes to the same register until the data from the previous write is synchronized to the AACIBITCLK domain. This takes two cycles of AACIBITCLK and two cycles of **PCLK**, and that is approximately 182ns at 100MHz **PCLK**.
- You do not program more than one channel to transmit data for a slot at a time. Similarly, you must not program more than one channel to receive data for a slot at a time.
- The ForcedRESET bit of the AACIRESET register is asserted LOW for at least 1µs in low-power mode in order to restart the AC-Link with a cold reset.
- The ForcedSYNC bit of the AACISYNC register must be kept asserted HIGH for at least 1.3µs in low-power mode in order to restart the AC-Link with a warm reset. The ForcedSYNC bit must not be set to 1 when the AACI is in normal mode.
- When the Transmit FIFO of a channel is in compact mode, the TSize bits in the corresponding AACITXCR1-4 register must indicate only a 12-bit, or 16-bit data width. Similarly, when the receive FIFO of a channel is in compact mode, the RSize bits in the corresponding AACIRXCR1-4 register must indicate only a 12-bit, or 16-bit data width.
- All slots programmed to be transmitted from, or received by a given channel must have the same sample rate.
- Data must be stored in the Tx FIFO lower slot first. For example, if slots 4 and 5 are to be stored in a FIFO, slot 4 must be the first data into the FIFO and slot 5 the second.
- The FIFOs must not be disabled in compact mode.

- The Slot Tx/Rx register enables (S11TxEn, S12TxEn, S112TxEn, S11RxEn, S12RxEn, S112RxEn bits in the AACIMAINCR register) must not be disabled while the corresponding Slot Tx/Rx busy bit of that register (S11TxBusy, S12TxBusy, S112TxBusy, S11RxBusy, S12RxBusy, S112RxBusy bits in the AACISLFR register) is HIGH.
- The AC-Link can be programmed into low-power mode only through the Slot Tx registers and not through the channels.
- All the receive logic and the FIFO transmit logic must be disabled and not busy before low-power mode is entered. The MainRxBusy bit of the AACIMAINFR register and the TxBusy bits of the AACISR1-4 registers must be polled for zero before programming the CODEC into low-power mode.
- The LowPower Mode bit in the AACIMAINCR register must be set only after the CODEC has been programmed to go to low-power mode, plus the time it takes for AACIBITCLK and AACISDATAIN to go LOW (normally the TS2_PDOWD parameter in the CODEC datasheets). If the LowPower Mode bit was set before this, the LOW to HIGH transitions on the AACISDATAIN line can result in the Wakeup interrupt being asserted. The Wakeup events on the AACISDATAIN line must be monitored once the LowPower Mode bit is set.
- Once a Slot Tx register has been written into and enabled, a new value must not be written into it until the S1TxEmpty flag corresponding to that register is asserted.
- Switching from non-compact mode to compact mode must not be done on the fly. The AACI must first be disabled before switching from non-compact mode to compact mode.
- Slot 2 data must be written first and then slot 1 data in the case of CODEC register write accesses. This is to prevent slot 1 data from being transmitted before the associated slot 2 data is available.
- If slots 1 and 2 are programmed in the same channel along with other slots, dummy data must be written for slot 2 during CODEC register read accesses. This is to prevent underrun.
- For CODEC register write accesses, if slots 1 and 2 are in separate channels, no slot other than slot 2 must be programmed in the same channel. In other words, slot 2 can be programmed into the same FIFO as slot 1 but not other slots.

———— **Note** ————

Slot 1 can be programmed with other slots in the same channel.

- SCRA bits in the AACIMAINCR register can be changed only when there is no data for slots 1 and 2 in the slot registers and the channels.

The LowPower Mode bit is set once the CODEC has been programmed to enter into low-power mode. (Power down register 0x26 has been programmable to the appropriate value through the AACISL1TX/AACISL2TX registers). You must ensure that this bit is only set after the CODEC has been programmed into low-power mode, plus the time it takes the **AACIBITCLK** and **AACISDATAIN** signals to go LOW (normally the TS2_PDOWN parameter in CODEC datasheets). You must ensure that the all receive logic and the FIFO transmit logic is disabled and not BUSY before low-power mode is entered. Once this bit has been set wake up events on the **AACISDATAIN** lines can be monitored.

You must ensure that the SCRA bits are changed only when there is no data for slots 1 and 2 in the slot registers and the channels. You must program SCRA bits before programming the register data (slot or channel), and this ensures that the correct CODEC is written or read.

3.7 Frequency restriction

The implementation of PrimeCell AACI limits the frequency ratios of **PCLK** to **AACIBITCLK** that are allowed for proper functioning of the PrimeCell AACI:

$$\mathbf{PCLK} \geq \frac{\mathbf{AACIBITCLK}}{5}$$

implies:

$$\mathbf{PCLK} \geq 2.4576\text{MHz}$$

Chapter 4

Programmer's Model for Test

This chapter describes the additional logic for functional verification and production testing. It contains the following sections:

- *PrimeCell AACI test harness overview* on page 4-2
- *Scan testing* on page 4-3
- *Test registers* on page 4-4
- *Integration testing of block inputs* on page 4-12
- *Integration testing of block outputs* on page 4-15
- *Integration test summary* on page 4-18.

4.1 PrimeCell AACI test harness overview

The additional logic for functional verification and integration vectors allows:

- stimulation of output signals to the block
- capture of the input signals.

The integration vectors provide a way of verifying that the PrimeCell AACI is correctly wired into a system. This is done by separately testing three groups of signals:

AMBA signals These are tested by checking the connections of all the address and data bits.

Primary input/output signals

These are tested using a simple trickbox that can demonstrate the correct connection of the input/output signals to external pads.

Intra-chip signals (such as interrupt sources)

The tests for these signals are system-specific, and enable you to write the necessary tests. Additional logic is implemented allowing you to read and write to each intra-chip input/output signal.

These test features are controlled by test registers. This allows you to test the PrimeCell AACI in isolation from the rest of the system, using only transfers from the AMBA APB.

Off-chip test vectors are supplied using a 32-bit parallel *External Bus Interface* (EBI) and converted to internal AMBA bus transfers. The application of test vectors is controlled through the *Test Interface Controller* (TIC) AMBA bus master module.

4.2 Scan testing

The PrimeCell AACI has been designed to simplify:

- insertion of scan test cells
- use of *Automatic Test Pattern Generation (ATPG)*.

This provides an alternative method of manufacturing test.

4.3 Test registers

The PrimeCell AACI test registers are memory-mapped as shown in Table 4-1.

Table 4-1 Test registers memory map

Address	Type	Width	Reset value	Name	Description
AACIBase + 0x180	Read/write	1	0x0	AACITCR	Test control register
AACIBase + 0x184	Read/write	3/2	0x0	AACIITIP	Integration test input register
AACIBase + 0x188	Read/write	30	0x00000000	AACIITOP0	Integration test output register 0
AACIBase + 0x18C	Read/write	13	0x0000	AACIITOP1	Integration test output register 1

Each register shown in Table 4-1 is described below.

4.3.1 Test control register, AACITCR

AACITCR is the test control register. This is a four-bit register. The ITEN bit forces the multiplexors on the inputs and outputs to use the test values. Table 4-2 shows the bit assignment for the AACITCR.

Table 4-2 AACITCR register

Bit	Name	Type	Description
31-4	-	-	Reserved.
3	REGTEST	Read/write	Register test bit: 0 = default, normal mode. Accesses to the registers are controlled by the hardware protection circuitry. 1 = test mode. The hardware protection circuitry is bypassed. You can perform normal write/read/write/read tests without dependence on the link side.

Table 4-2 AACITCR register (continued)

Bit	Name	Type	Description
2-1	FIFOTEST	Read/write	FIFO test bits: 00 = default, normal mode. Reads to the data register return data from the read port of the Rx FIFO. Writes to the data register write data into the write port of the Tx FIFO. 01 = test mode. Reads to the data register return data from the Read port of the Tx FIFO. (The read to the read port of the Rx FIFO is suppressed.) Writes to the data register write data into the write port of the Rx FIFO. (The write to the write port of the Tx FIFO is suppressed.) 10 = test mode. Reads to the data register return data from the read port of the Rx FIFO. This read access automatically generates a write access to the write port of the Rx FIFO. Writes to the data register write data into the write port of the Tx FIFO. ^a 11 = test mode. Reads to the data register return data from the Read port of the Tx FIFO. This read access automatically generates a write access to the write port of the Tx FIFO. Writes to the data register write data into the write port of the Rx FIFO. ^b
0	ITEN	Read/write	Integration test enable: 0 = PrimeCell AACI placed in normal mode 1 = PrimeCell AACI placed in integration test mode.

a. The actual test code does not perform writes when FIFOTEST = 10.

b. The actual test code does not perform writes when FIFOTEST = 11.

4.3.2 Integration test output register, AACIITIP

AACIITIP is the integration test output register. AACIITIP is a 3-bit register. Out of these three bits, only the AACIDMACLRRX and AACIDMACLRTX bits are writable. AACISDATAIN is read-only. Table 4-3 shows the bit assignments for the AACIITIP.

Table 4-3 AACIITIP register

Bit	Name	Type	Description
31-3	-	-	Reserved
2	AACISDATAIN	Read	Reads return the value on the AACISDATAIN primary input.

Table 4-3 AACIITIP register (continued)

Bit	Name	Type	Description
1	AACIDMACLRTX	Read/write	Writes specify the value to be driven on the intra-chip AACIDMACLRTX input in the integration test mode. Reads return the value on AACIDMACLRTX at the output of the test multiplexor.
0	AACIDMACLRRX	Read/write	Writes specify the value to be driven on the intra-chip AACIDMACLRRX input in the integration test mode. Reads return the value on AACIDMACLRRX at the output of the test multiplexor.

4.3.3 Integration test output register, AACIITOP0

AACIITOP0 is the integration test output set or read register 0. Table 4-4 shows the bit assignment for the AACIITOP0.

Table 4-4 AACIITOP0 register

Bit	Name	Type	Description
31-30	-		Reserved.
29	AACISDATAOUT	Read/write	Primary output. Writes specify the value to be driven on the AACISDATAOUT primary output in the integration test mode. Reads return the value written into this field.
28	AACIDMABREQTX	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIDMABREQTX output in the integration test mode. Reads return the value on AACIDMABREQTX at the output of the test multiplexor.
27	AACIDMASREQRX	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIDMASREQRX output in the integration test mode. Reads return the value on AACIDMASREQRX at the output of the test multiplexor.
26	AACIDMABREQRX	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIDMABREQRX output in the integration test mode. Reads return the value on AACIDMABREQRX at the output of the test multiplexor.

Table 4-4 AACIITOP0 register (continued)

Bit	Name	Type	Description
25	AACIDMALSREQRX	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIDMALSREQRX output in the integration test mode. Reads return the value on AACIDMALSREQRX at the output of the test multiplexor.
24	AACIDMALBREQRX	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIDMALBREQRX output in the integration test mode. Reads return the value on AACIDMALBREQRX at the output of the test multiplexor.
23	AACIRXINTR1	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXINTR1 output in the integration test mode. Reads return the value on AACIRXINTR1 at the output of the test multiplexor.
22	AACIRXINTR2	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXINTR2 output in the integration test mode. Reads return the value on AACIRXINTR2 at the output of the test multiplexor.
21	AACIRXINTR3	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXINTR3 output in the integration test mode. Reads return the value on AACIRXINTR3 at the output of the test multiplexor.
20	AACIRXINTR4	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXINTR4 output in the integration test mode. Reads return the value on AACIRXINTR4 at the output of the test multiplexor.
19	AACITXINTR1	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXINTR1 output in the integration test mode. Reads return the value on AACITXINTR1 at the output of the test multiplexor.
18	AACITXINTR2	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXINTR2 output in the integration test mode. Reads return the value on AACITXINTR2 at the output of the test multiplexor.

Table 4-4 AACIITOP0 register (continued)

Bit	Name	Type	Description
17	AACITXINTR3	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXINTR3 output in the integration test mode. Reads return the value on AACITXINTR3 at the output of the test multiplexor.
16	AACITXINTR4	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXINTR4 output in the integration test mode. Reads return the value on AACITXINTR4 at the output of the test multiplexor.
15	AACIORINTR1	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIORINTR1 output in the integration test mode. Reads return the value on AACIORINTR1 at the output of the test multiplexor.
14	AACIORINTR2	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIORINTR2 output in the integration test mode. Reads return the value on AACIORINTR2 at the output of the test multiplexor
13	AACIORINTR3	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIORINTR3 output in the integration test mode. Reads return the value on AACIORINTR3 at the output of the test multiplexor
12	AACIORINTR4	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIORINTR4 output in the integration test mode. Reads return the value on AACIORINTR4 at the output of the test multiplexor
11	AACIURINTR1	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIURINTR1 output in the integration test mode. Reads return the value on AACIURINTR1 at the output of the test multiplexor.
10	AACIURINTR2	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIURINTR2 output in the integration test mode. Reads return the value on AACIURINTR2 at the output of the test multiplexor.
9	AACIURINTR3	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIURINTR3 output in the integration test mode. Reads return the value on AACIURINTR3 at the output of the test multiplexor.

Table 4-4 AACIITOP0 register (continued)

Bit	Name	Type	Description
8	AACIURINTR4	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIURINTR4 output in the integration test mode. Reads return the value on AACIURINTR4 at the output of the test multiplexor.
7	AACIRXTOINTR1	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOINTR1 output in the integration test mode. Reads return the value on AACIRXTOINTR1 at the output of the test multiplexor.
6	AACIRXTOINTR2	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOINTR2 output in the integration test mode. Reads return the value on AACIRXTOINTR2 at the output of the test multiplexor.
5	AACIRXTOINTR3	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOINTR3 output in the integration test mode. Reads return the value on AACIRXTOINTR3 at the output of the test multiplexor.
4	AACIRXTOINTR4	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOINTR4 output in the integration test mode. Reads return the value on AACIRXTOINTR4 at the output of the test multiplexor.
3	AACITXCINTR1	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXCINTR1 output in the integration test mode. Reads return the value on AACITXCINTR1 at the output of the test multiplexor.
2	AACITXCINTR2	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXCINTR2 output in the integration test mode. Reads return the value on AACITXCINTR2 at the output of the test multiplexor.
1	AACITXCINTR3	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXCINTR3 output in the integration test mode. Reads return the value on AACITXCINTR3 at the output of the test multiplexor.
0	AACITXCINTR4	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACITXCINTR4 output in the integration test mode. Reads return the value on AACITXCINTR4 at the output of the test multiplexor.

4.3.4 Integration test output register, AACIITOP1

AACIITOP1 is the integration test output set or read register 1. Table 4-5 shows the bit assignment for the AACIITOP1.

Table 4-5 AACIITOP1 register

Bit	Name	Type	Description
12	AACIRXTOFEINTR1	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOFEINTR1 output in the integration test mode. Reads return the value on AACIRXTOFEINTR1 at the output of the test multiplexor.
11	AACIRXTOFEINTR2	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOFEINTR2 output in the integration test mode. Reads return the value on AACIRXTOFEINTR2 at the output of the test multiplexor.
10	AACIRXTOFEINTR3	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOFEINTR3 output in the integration test mode. Reads return the value on AACIRXTOFEINTR3 at the output of the test multiplexor.
9	AACIRXTOFEINTR4	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIRXTOFEINTR4 output in the integration test mode. Reads return the value on AACIRXTOFEINTR4 at the output of the test multiplexor.
8	AACIWINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIWINTR output in the integration test mode. Reads return the value on AACIWINTR at the output of the test multiplexor.
7	AACIGPIOINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIGPIOINTR output in the integration test mode. Reads return the value on AACIGPIOINTR at the output of the test multiplexor.
6	AACIS12RXINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIS12RXINTR output in the integration test mode. Reads return the value on AACIS12RXINTR at the output of the test multiplexor.

Table 4-5 AACIITOP1 register (continued)

Bit	Name	Type	Description
5	AACIS12TXINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIS12TXINTR output in the integration test mode. Reads return the value on AACIS12TXINTR at the output of the test multiplexor.
4	AACIS2RXINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIS2RXINTR output in the integration test mode. Reads return the value on AACIS2RXINTR at the output of the test multiplexor.
3	AACIS2TXINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIS2TXINTR output in the integration test mode. Reads return the value on AACIS2TXINTR at the output of the test multiplexor.
2	AACIS1RXINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIS1RXINTR output in the integration test mode. Reads return the value on AACIS1RXINTR at the output of the test multiplexor.
1	AACIS1TXINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIS1TXINTR output in the integration test mode. Reads return the value on AACIS1TXINTR at the output of the test multiplexor.
0	AACIINTR	Read/write	Intra-chip output. Writes specify the value to be driven on the intra-chip AACIINTR output in the integration test mode. Reads return the value on AACIINTR at the output of the test multiplexor.

AACIITOP0 is a 30-bit register and AACIITOP1 is a 13-bit register. AACIITOP0[28:0] and AACIITOP1[12:0] are used in the integration testing of intra-chip outputs and AACIITOP[29] is used in the integration testing of primary outputs. The read path is different for the intra-chip and the primary output pins. In the case of primary outputs, reads return the values that have been written to the corresponding bit in the AACIITOP0 register. In the case of intra-chip outputs, if ITEN bit is set, AACITOP0[28:0] and AACIITOP1[12:0] are connected with the actual outputs. Otherwise these pins are connected with the stored register values.

4.4 Integration testing of block inputs

The following sections describe the integration testing for the block inputs:

- *Intra-chip inputs*
- *Primary inputs* on page 4-13.

4.4.1 Intra-chip inputs

When you run integration tests with the PrimeCell AACI in a standalone test setup:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the AACIITIP[1:0] register bits to the **AACIDMACLRRX** and **AACIDMACLRTX** signals.
- Write a 1 and then a 0 to each of the AACIITIP[1:0] register bits, and read the same register bits to ensure that the value written is read out.

When you run integration tests with the PrimeCell AACI as part of an integrated system:

- Write a 0 to the ITEN bit in the control register. This selects the normal path from the external **AACIDMACLRRX** pin to the internal **AACIDMACLRRX** signal, and the path from the external **AACIDMACLRTX** pin to the internal **AACIDMACLRTX** pin.
- Write a 1 and then a 0 to the internal test registers of the DMA controller to toggle the **AACIDMACLRRX** signal connection between the DMA controller and the PrimeCell AACI. Read from the AACIITIP[0] register bit to verify that the value written into the DMA controller, is read out through the PrimeCell AACI. Similarly, write a 1 and then a 0 to the internal registers of the DMA controller to toggle the **AACIDMACLRTX** signal connection between the DMA controller and the PrimeCell AACI. Read from the AACIITIP[1] register bit to verify that the value written into the DMA controller, is read out through the PrimeCell AACI.

Figure 4-1 on page 4-13 shows the implementation details of the input integration test harness.

The ITEN bit is used as the control bit for the multiplexor in the read path of the AACIDMACLRRX and AACIDMACLRTX intra-chip inputs. If the ITEN control bit is deasserted, the AACIDMACLRRX and AACIDMACLRTX intra-chip inputs are

routed as the internal AACIDMACLRRX and AACIDMACLRTX inputs respectively, otherwise the stored register values are driven on the internal line. The read-only bit AACISDATAIN is connected directly to the primary input pin.

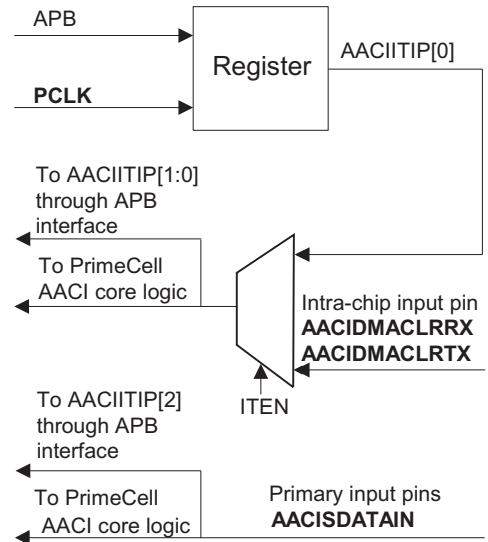


Figure 4-1 Input integration test harness

4.4.2 Primary inputs

Use this test for the following input:

- **AACISDATAIN.**

AACISDATAIN is tested using the integration vector trickbox by implementing a simple OR tree loop back. All the primary output lines (**AACISDATAOUT**, **AACISYNC**, and **AACIRESET**) are OR-ed together and routed to **AACISDATAIN**. The exact test sequence to verify the connectivity of **AACISDATAIN** is detailed in *Primary outputs* on page 4-16.

The primary input **AACIBITCLK** is tested indirectly by connecting a free-running clock generated in the integration vector trickbox to the **AACIBITCLK** input pin of the PrimeCell AACI. The exact test sequence is as follows:

- Assert and negate the intra-chip input, **nAACIBITCLKRST** through the reset controller. This resets all flip-flops in the PrimeCell AACI that operate on **AACIBITCLK**. This indirectly tests the connectivity of the **nAACIBITCLKRST** intra-chip input from the reset controller to the PrimeCell AACI.

- Configure the AACITXCR registers for 0 slots. Write some data into the AACISL1TX register, and write a 1 to the Sl1TxEn bit of the AACIMAINCR register. Write a 1 to AACITOP0[29], AACIRESET and the AACISYNC register to make the AACISDATAIN bit of the AACITIP a 1. This simulates a *CODEC Ready* to allow the reception of a frame. Write a 1 into the ITEN bit of the integration test control register to select the integration test mode paths in the multiplexers on the primary outputs. This forces **AACISDATAIN HIGH** through the integration vector trickbox.
- Write a 1 to the AacIfE bit of the AACIMAINCR register. After waiting for enough time to allow slot 1 data to be received by the PrimeCell AACI, the Sl1TxEmpty bit of the AACISLFR register is polled. If this bit is set, it proves connectivity of the **AACIBITCLK** pin as well as the **nAACIBITCLKRST** intra-chip input.

4.5 Integration testing of block outputs

The following sections describe the integration testing for the block outputs:

- *Intra-chip outputs*
- *Primary outputs* on page 4-16.

4.5.1 Intra-chip outputs

Use this test for the following outputs:

- **AACIDMASREQRX**
- **AACIDMALSREQRX**
- **AACIDMABREQRX**
- **AACIDMALBREQRX**
- **AACIINTR**
- **AACIRXINTR1-4**
- **AACITXINTR1-4**
- **AACIORINTR1-4**
- **AACIURINTR1-4**
- **AACIRXTOINTR1-4**
- **AACITXCINTR1-4**
- **AACIWINTR**
- **AACIGPIOINTR**
- **AACIS12RXINTR**
- **AACIS12TXINTR**
- **AACIS2RXINTR**
- **AACIS2TXINTR**
- **AACIS1RXINTR**
- **AACIS1TXINTR**
- **AACIDMABREQTX**
- **AACIRXTOFEINTR1**
- **AACIRXTOFEINTR2**
- **AACIRXTOFEINTR3**
- **AACIRXTOFEINTR4**
- **AACIINTR.**

When you run integration tests with the PrimeCell AACI in a standalone test setup:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the AACIITOP0[28:0] and AACIITOP1[12:0] register bits to the intra-chip output signals.

- Write a 1 and then a 0 to the AACIITOP0[28:0] and AACIITOP1[12:0] register bits, and read the same register bits to verify that the value written is read out.

When you run integration tests with the PrimeCell AACI as part of an integrated system:

- Write a 0 to the ITEN bit in the control register. This selects the test path from the AACIITOP0[28:0] and AACIITOP1[12:0] register bits to the intra-chip output signals.
- Write a 1 and then a 0 to the AACIITOP0[28:0] and AACIITOP1[12:0] register bits to toggle the signal connections between the DMA controller/interrupt controller and the PrimeCell AACI. Read from the internal test registers of the DMA controller/interrupt controller to verify that the value written into the AACIITOP0[28:0] and AACIITOP1[12:0] register bits is read out through the PrimeCell AACI.

Figure 4-2 shows details of the implementation of the output integration test harness in the case of intra-chip outputs.

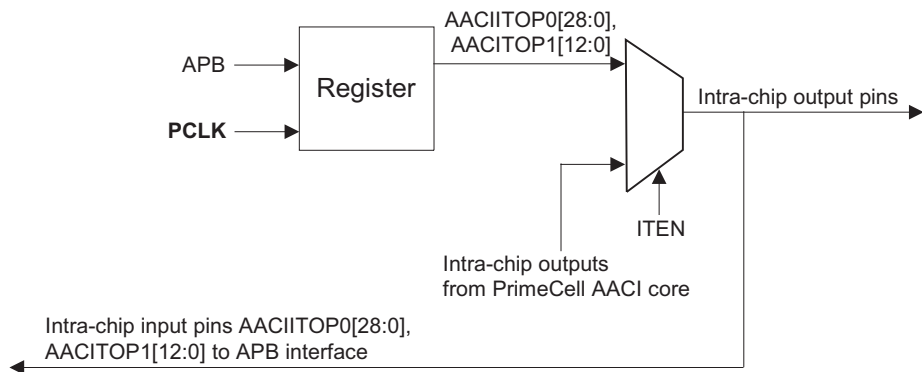


Figure 4-2 Output integration test harness, intra-chip outputs

4.5.2 Primary outputs

Integration testing of primary outputs and primary inputs is carried out using the integration vector trickbox. Use this test for the following outputs:

- **AACISDATAOUT**
- **AACIRESET**
- **AACISYNC.**

Verify the **AACISDATAOUT**, **AACIRESET**, and **AACISYNC** primary input/output pin connections as follows:

- Primary output pins, **AACISDATAOUT**, **AACIRESET**, and **AACISYNC** are ORed together and routed back to the primary input pin **AACISDATAIN** through the integration vector trickbox.
- All the primary outputs can be accessed through the AACIITOP0 register. Different data patterns are written to the output pins using the AACIITOP0 registers.
- The ORed data is read back from the **AACISDATAIN** pin through the AACIITIP register.

Figure 4-3 shows details of the implementation of the output integration test harness in the case of primary outputs.

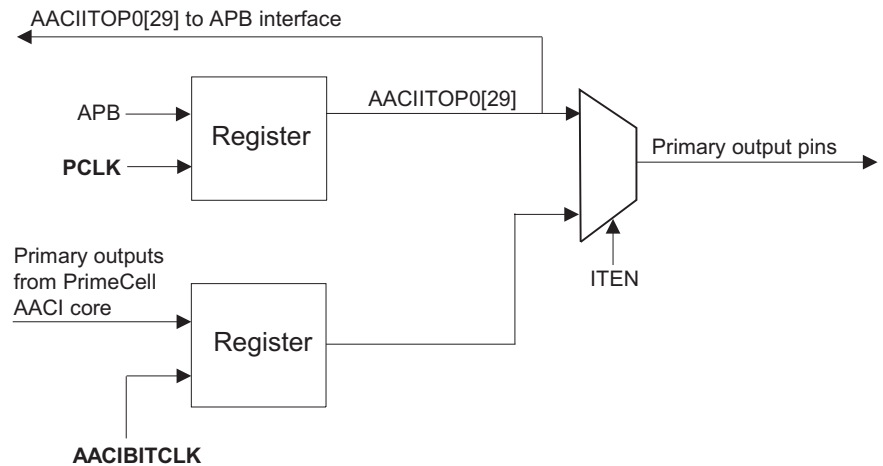


Figure 4-3 Output integration test harness, primary outputs

4.6 Integration test summary

Table 4-6 summarizes the integration test strategy for all PrimeCell AACI pins.

Table 4-6 PrimeCell AACI integration test strategy

Name	Type	Source/ destination	Test strategy
PCLK	In	APB	Register read/write.
PRESETn	In	APB	Register read/write.
PADDR[11:2]	In	APB	Register read/write.
PWDATA[31:0]	In	APB	Register read/write.
PRDATA[31:0]	Out	APB	Register read/write.
PENABLE	In	APB	Register read/write.
PWRITE	In	APB	Register read/write.
PSEL	In	APB	Register read/write.
AACIBITCLK	In	Primary	Indirectly tested by performing a data transfer.
nAACIBITCLK	In	Clock generator	Not tested using integration test vectors.
AACISDATAOUT	Out	Primary	Use integration vector trickbox, and AACIITIP, AACIRESET, AACISYNC, and AACIITOP registers.
AACISDATAIN	In	Primary	Use integration vector trickbox, and AACIITIP, AACIRESET, AACISYNC, and AACIITOP registers.
SCANENABLE	In	Test controller	Not tested using integration test vectors.
SCANINPCLK	In	Test controller	Not tested using integration test vectors.
SCANINnBITCLK	In	Test controller	Not tested using integration test vectors.
SCANINBITCLK	In	Test controller	Not tested using integration test vectors.
SCANOUTPCLK	Out	Test controller	Not tested using integration test vectors.
SCANOUTBITCLK	Out	Test controller	Not tested using integration test vectors.
SCANOUTnBITCLK	Out	Test controller	Not tested using integration test vectors.
AACIINTR	Out	Intra-chip	Use AACIITOP1 register.

Table 4-6 PrimeCell AACI integration test strategy (continued)

Name	Type	Source/ destination	Test strategy
AACIRXINTR1	Out	Intra-chip	Use AACIITOP0 register.
AACIRXINTR2	Out	Intra-chip	Use AACIITOP0 register.
AACIRXINTR3	Out	Intra-chip	Use AACIITOP0 register.
AACIRXINTR4	Out	Intra-chip	Use AACIITOP0 register.
AACITXINTR1	Out	Intra-chip	Use AACIITOP0 register.
AACITXINTR2	Out	Intra-chip	Use AACIITOP0 register.
AACITXINTR3	Out	Intra-chip	Use AACIITOP0 register.
AACITXINTR4	Out	Intra-chip	Use AACIITOP0 register.
AACIRXTOINTR1	Out	Intra-chip	Use AACIITOP0 register.
AACIRXTOINTR2	Out	Intra-chip	Use AACIITOP0 register.
AACIRXTOINTR3	Out	Intra-chip	Use AACIITOP0 register.
AACIRXTOINTR4	Out	Intra-chip	Use AACIITOP0 register.
AACITXCINTR1	Out	Intra-chip	Use AACIITOP0 register.
AACITXCINTR2	Out	Intra-chip	Use AACIITOP0 register.
AACITXCINTR3	Out	Intra-chip	Use AACIITOP0 register.
AACITXCINTR4	Out	Intra-chip	Use AACIITOP0 register.
AACIURINTR1	Out	Intra-chip	Use AACIITOP0 register.
AACIURINTR2	Out	Intra-chip	Use AACIITOP0 register.
AACIURINTR3	Out	Intra-chip	Use AACIITOP0 register.
AACIURINTR4	Out	Intra-chip	Use AACIITOP0 register.
AACIORINTR1	Out	Intra-chip	Use AACIITOP0 register.
AACIORINTR2	Out	Intra-chip	Use AACIITOP0 register.
AACIORINTR3	Out	Intra-chip	Use AACIITOP0 register.

Table 4-6 PrimeCell AACI integration test strategy (continued)

Name	Type	Source/ destination	Test strategy
AACIORINTR4	Out	Intra-chip	Use AACIITOP0 register.
AACIWINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIGPIOINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIS1RXINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIS1TXINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIS2RXINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIS2TXINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIS12RXINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIS12TXINTR	Out	Intra-chip	Use AACIITOP1 register.
AACIDMASREQRX	Out	Intra-chip	Use AACIITOP0 register.
AACIDMALSREQRX	Out	Intra-chip	Use AACIITOP0 register.
AACIDMABREQRX	Out	Intra-chip	Use AACIITOP0 register.
AACIDMALBREQRX	Out	Intra-chip	Use AACIITOP0 register.
AACIDMACLRRX	In	Intra-chip	Use AACIITIP register.
AACIDMABREQTX	Out	Intra-chip	Use AACIITOP0 register.
AACIDMACLRTX	In	Intra-chip	Use AACIITIP register.
AACISYNC	Out	Primary	Use integration vector trickbox, AACITIP, AACITOP, AACISYNC and AACIRESET registers.
AACIRESET	Out	Primary	Use integration vector trickbox, AACITIP, AACITOP, AACISYNC and AACIRESET registers.
AACIRXTOFEINTR1	Out	Intra-chip	Use AACIITOP1 register.
AACIRXTOFEINTR2	Out	Intra-chip	Use AACIITOP1 register.
AACIRXTOFEINTR3	Out	Intra-chip	Use AACIITOP1 register.
AACIRXTOFEINTR4	Out	Intra-chip	Use AACIITOP1 register.

Note

nAACIBITCLKRST is implicitly tested through integration methodology. At least one reset value test will fail if **nAACIBITCLKRST** is not applied properly.

Appendix A

ARM PrimeCell AACI (PL041) Signal Descriptions

This appendix describes the signals which interface with the PrimeCell Advanced Audio CODEC Interface (PL041) block. It contains the following:

- *AMBA APB signals* on page A-2
- *Module-specific signals* on page A-3.

A.1 AMBA APB signals

The PrimeCell AACI module is connected to the AMBA APB as a bus slave. The AMBA APB signals have a **P** prefix and are active HIGH. Active LOW signals contain a lower case **n**. The AMBA APB signals are described in Table A-1.

Table A-1 AMBA APB signal descriptions

Name	Type	Source/ destination	Description
PCLK	Input	Clock generator	AMBA APB clock.
PRESETn	Input	Reset controller	Bus reset signal, active LOW.
PADDR [11:2]	Input	APB bridge	Subset of AMBA APB address bus.
PWDATA [31:0]	Input	APB bridge	Data is written into the peripheral on this bus.
PRDATA [31:0]	Output	APB data multiplexor	Data is read out of the peripheral on this bus.
PENABLE	Input	APB bridge	This signal is used to time all accesses on the peripheral bus.
PWRITE	Input	APB bridge	When HIGH this signal indicates a write into the peripheral, when LOW it indicates a read from the peripheral.
PSEL	Input	APB bridge	When HIGH this signal indicates the peripheral is selected by the AMBA APB bridge.

A.2 Module-specific signals

Table A-2 describes the module-specific signals.

Table A-2 Module specific signal descriptions

Name	Type	Source/ destination	Description
AACIBITCLK	Input	Audio CODEC	Clock supplied by the audio CODEC. Fixed at 12.288MHz.
nAACIBITCLK	Input	Clock generator	Inverted AACIBITCLK . Fixed at 12.288MHz.
nAACIBITCLKRST	Input	Reset controller	Reset signal to the AACIBITCLK clock domain.
AACISYNC	Output	Audio CODEC	Frame synchronization output pulse. Fixed at 48kHz, derived by dividing down the serial clock AACIBITCLK by 256 and is set from the rising edge of AACIBITCLK . The AACISYNC is also output asynchronously when the Audio CODEC is in the WARM RESET state.
AACIRESET	Output	Audio CODEC	Reset signal from the AACI to the off-chip audio CODEC. Asynchronous cold reset (active LOW, reset registers).
AACISDATAIN	Input	Audio CODEC	Serial data input from off-chip audio CODEC. This input receives the status information and digital audio input streams. Sampled on the falling edge of AACIBITCLK , it is output from the CODEC on the rising edge. When there is more than one CODEC, all AACISDATAIN lines must be connected through an external OR gate and input to AACISDATAIN .
AACISDATAOUT	Output	Audio CODEC	Serial data output to off-chip audio CODEC. This output transmits the control information and digital audio output streams. Output on the rising edge of AACIBITCLK
SCANENABLE	Input	Test controller	Scan enable signal for all clock domains.
SCANINPCLK	Input	Test controller	Scan input signal for PCLK domain.
SCANINBITCLK	Input	Test controller	Scan input signal for AACIBITCLK domain.
SCANINnBITCLK	Input	Test controller	Scan input signal for nAACIBITCLK domain.
SCANOUTPCLK	Output	Test controller	Scan output signal for PCLK domain.
SCANOUTBITCLK	Output	Test controller	Scan output signal for AACIBITCLK domain.

Table A-2 Module specific signal descriptions (continued)

Name	Type	Source/ destination	Description
SCANOUTnBITCLK	Output	Test controller	Scan output signal for nAACIBITCLK domain.
AACIINTR	Output	Interrupt controller	Combined AACI interrupt. Asserted when any of the individual interrupts are asserted.
AACIRXINTR1-4	Output	Interrupt controller	Interrupts of the receive FIFOs. Asserted when they are at least half full.
AACITXINTR1-4	Output	Interrupt controller	Interrupts of the transmit FIFOs. Asserted when they are at least half full.
AACIRXTOINTR1-4	Output	Interrupt controller	Receive timeout interrupts. Asserted when the receive FIFOs are not empty and no further data is received for the TOC number of frames programmed in the AACIRXCR registers.
AACITXCINTR1-4	Output	Interrupt controller	Transmit complete interrupts. Asserted when transmit FIFOs are empty and transmit shift register is empty.
AACIURINTR1-4	Output	Interrupt controller	Transmit FIFOs underrun interrupt. Asserted when data for all slots programmed in the AACITXCR registers is not present in transmit FIFOs at the time of transmission.
AACIORINTR1-4	Output	Interrupt controller	Receive FIFO overrun interrupts. Asserted if data is received and receive FIFOs are already full.
AACIWINTR	Output	Interrupt controller	Wake up interrupt. Asserted by a HIGH on AACISDATAIN in the absence of AACIBITCLK .
AACIGPIOINTR	Output	Interrupt controller	GPIO interrupt. Asserted when bit 0 of slot 12 of the incoming AACISDATAIN is 1.
AACIS1RXINTR	Output	Interrupt controller	Slot 1 Rx interrupt. Asserted when AACISL1RX register has new data that is not read.
AACIS1TXINTR	Output	Interrupt controller	Slot 1 Tx interrupt. Asserted when there is no data present in AACISL1TX register.
AACIS2RXINTR	Output	Interrupt controller	Slot 2 Rx interrupt. Asserted when AACISL2RX register has new data that is not read.
AACIS2TXINTR	Output	Interrupt controller	Slot 2 Tx interrupt. Asserted when there is no data present in AACISL2TX register.
AACIS12RXINTR	Output	Interrupt controller	Slot 12 Rx interrupt. Asserted when AACISL12RX register has new data that is not read.

Table A-2 Module specific signal descriptions (continued)

Name	Type	Source/ destination	Description
AACIS12TXINTR	Output	Interrupt controller	Slot 12 Tx interrupt. Asserted when there is no data present in AACISL12TX register.
AACIRXTOFEINTR1-4	Output	Interrupt controller	Receive timeout FIFO empty interrupts. Asserted when receive FIFOs are empty and no further data is received for the TOC number of frames programmed in the AACIRXCR registers.
AACIDMASREQRX	Output	DMA controller	Single-word DMA transfer request, asserted by the PrimeCell AACI. For receive, the signal is asserted if the receive FIFO contains between one and three words and RxTimeout has not been generated. AACIDMASREQRX assertion: With timeout it is asserted if FIFO fill level is equal to two or three. Without timeout it is asserted if FIFO fill level is equal to one, two or three.
AACIDMALSREQRX	Output	DMA controller	Last single-word DMA transfer request, asserted by the PrimeCell AACI. For receive, the signal is asserted if the RxTimeout bit in the AACISR is valid and the FIFO contains only one word.
AACIDMABREQRX	Output	DMA controller	Burst DMA transfer request, asserted by the PrimeCell AACI. For receive, the signal is asserted if the FIFO contains four or more words. AACIDMASREQRX assertion: With timeout it is asserted if FIFO fill level is greater than four. Without timeout it is asserted if FIFO fill level is greater than, or equal to four.
AACIDMALBREQRX	Output	DMA controller	Last burst DMA transfer request, asserted by the PrimeCell AACI. For receive, the signal is asserted if the RxTimeout bit in the AACISR is valid and the FIFO contains only four words.

Table A-2 Module specific signal descriptions (continued)

Name	Type	Source/ destination	Description
AACIDMACLRRX	Input	DMA controller	DMA request clear, asserted by the DMAC to clear the request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.
AACIDMABREQTX	Output	DMA controller	Burst DMA transfer request, asserted by the PrimeCell AACI. For transmit, the signal is asserted if the FIFO contains four or less words.
AACIDMACLRTX	Input	DMA controller	DMA request clear, asserted by the DMAC to clear the request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

Appendix B

Frame format

This appendix describes the frame format of the **AACISDATAOUT** and **AACISDATAIN** signals. It contains the following:

- *AACISDATAOUT frame format* on page B-2
- *AACISDATAIN frame format* on page B-5.

B.1 AACISDATAOUT frame format

The information transmitted on **AACISDATAOUT** is made up of a number of different slots. Each slot is made up of a number of bits and each new bit position is presented to the AC-LINK on a rising edge of **AACIBITCLK**.

Figure B-1 shows the make-up of a bidirectional audio frame. The frame is made up of a tag and a data phase each with slot(s). Slot 0 is the tag phase and slots 1 to 12 are the data phase. The data phase slots are each allocated to a different function.

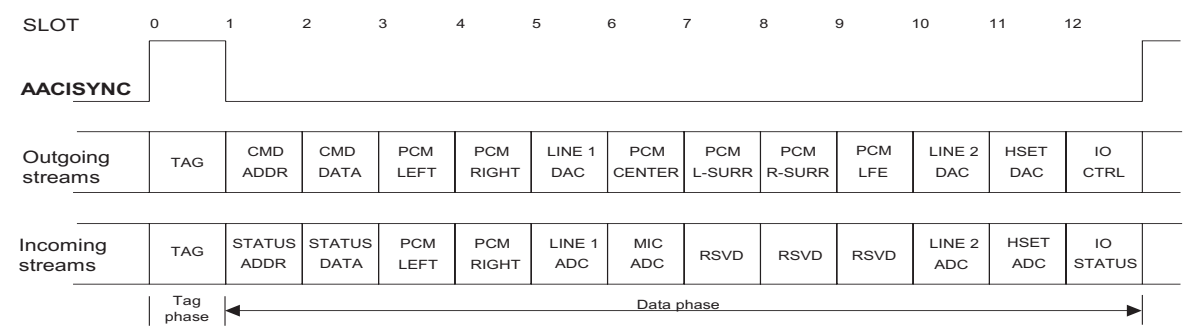


Figure B-1 Bidirectional audio frame

The slots are described in Table B-1 on page B-3.

Table B-1 AACISDATAOUT slot description

Slot	Type	Bits	Description
0	Tag phase	16	The first bit of slot 0 is designated the <i>valid frame</i> bit. If this first bit is 1, it indicates that the current data frame contains at least one slot of valid data and the external CODEC must sample the following bits in this slot 0 to determine which frames do in fact have valid data. Valid slots are signified by a 1 in their respective bit position. If a slot is not valid the PrimeCell AACI fills the relevant AACISDATAOUT slots with zeros. For a read operation, slot 1 valid is set when the register address (slot 1) contains valid data. For a write operation, slot 1 and slot 2 valid is set to indicate that the register write data (slot 2) contains data. When bits SCRA[1:0] of AACIMAINCR are not zero (secondary CODEC), bits 13 and 14 of slot 0 must be zero, for both the read and write operation. The SCRA bits only indicate which CODEC is being addressed by the PrimeCell AACI for register access. The SCRA bits allow you to specify which CODEC the slot 1 and 2 data is for. All other slot data is transmitted as normal and the external CODECs determine which slot they require. Slot 0 is determined by the values in the AACITXCR registers and if the FIFO contains data. The TXSLOT data from each AACITXCR register is qualified by the valid bits of the FIFO and then ORed together with the qualified TXSLOT data from the other channels. The AACISL1TX, AACISL2TX, and AACISL12TX registers are also monitored. If any new data is written into the registers the appropriate bit in slot 0 is set.
1	Register address	20	This slot indicates the register address of the register access for the current frame. The <i>Most Significant Bit</i> (MSB) of slot 1 (bit19) signifies whether the current control operation is a read (0) or a write (1). Bits 18-12 are used to specify the register address of the read or write operation. The trailing 12 bits are reserved and must be zero (this is under software control).
2	Register write data	20	This slot contains the register data of the register write access for the current frame (as defined in slot 1). Bits 19-4 contain the 16-bit value to be written to the register. The hardware zeros bits 3-0. If the access is a register read, the entire slot fills with zeros (this requirement is determined by the PrimeCell AACI that uses bit 19 in slot 1 to determine if the operation is a read or write).
3	PCM playback left channel	20	This slot transmits PCM playback data intended for the left channel DAC on the CODEC.
4	PCM playback right channel	20	This slot transmits PCM playback data intended for the right channel DAC on the CODEC.
5	Modem 1 channel	20	This slot transmits data intended for modem line 1 DAC on the CODEC.

Table B-1 AACISDATAOUT slot description (continued)

Slot	Type	Bits	Description
6	PCM playback centre channel	20	This slot transmits PCM playback data intended for the centre channel on the CODEC.
7	PCM playback left surround channel	20	This slot transmits PCM playback data intended for the left surround channel on the CODEC.
8	PCM playback right surround channel	20	This slot transmits PCM playback data intended for the right surround channel on the CODEC.
9	PCM playback low frequency effects surround channel.	20	This slot transmits PCM playback data intended for the <i>Low Frequency Effects</i> (LFE) channel on the CODEC.
10	Modem 2 channel	20	This slot can either transmit data intended for modem line two DAC or PCM left channel when 96kHz sampling rate is required (for example for DVD players).
11	Headset channel	20	This slot can transmit data intended for headset or PCM right channel when 96kHz sampling rate is required (for example for DVD players).
12	I/O control channel	20	This slot can transmit data intended for I/O control or PCM centre channel when 96kHz sampling rate is required (for example for DVD players).

B.2 AACISDATAIN frame format

The information transmitted on **AACISDATAIN** is made up of a number of different slots. Each slot is made up of a number of bits and each new bit position is presented to the AC-link on a rising edge of **AACIBITCLK**.

The slots are described in Table B-2.

Table B-2 AACISDATAIN slot description

Slot	Type	Bits	Description
0	CODEC status bits	16	The first bit of slot 0 (bit 15) indicates when the CODEC is ready. The PrimeCell AACI checks the following data bits in slot 0 to see which other subsections are ready. For example, a 1 in bit 14 indicates that the data in slot 1 is valid and a 1 in bit 13 indicates that the data in slot 2 is valid. The data in this slot is stored in the PrimeCell AACI in order to allow bits from this slot to validate the slots within the frame.
1	Status address	20	This slot echoes the register address of the external CODEC back to the PrimeCell AACI when the CODEC is issued a read request from the previous frame. The external CODEC only echoes the register index for a read access. Write accesses do not return valid data in slot 1. For reads, bit 19 is always a zero. Bits 18-12 contain the 7-bit register index for the CODEC registers. Bits 11-2 are used for <i>Sample Rate Conversion</i> (SRC) functionality, known as disable request bits for slots 3-12. When the external CODEC sets these bits to 1 the corresponding slot must not send data on AACISDATAOUT in the following frame. When the external CODEC sets these bits to 0 the corresponding slot must respond with data on AACISDATAOUT in the next frame. The disable request bits have no effect for a particular slot if the controller has not been set up to store that slot and zeros are transmitted in the next frame. If the channel (set up to store the requested slot) has no data available for transmission, an underflow occurs and invalid data, all zeros, is transmitted from the FIFO until the FIFO becomes non-empty or until transmit is disabled. The system containing the PrimeCell AACI must use the transmit interrupt request to write transmit data at a sufficient rate to prevent an underflow error condition. The disable request bits are independent of the slot 0 tag bit for slot 1, the tag bit is only set when slot 1 is returning a valid address.
2	Status data	20	This slot returns the control register data requested by the PrimeCell AACI from the previous read request. If slot 2 is tagged as being invalid by the CODEC status bits in slot 0 then the entire slot is filled with zeros by the CODEC. Bits 19-4 contain the 16-bit CODEC register value returned to the PrimeCell AACI. Bits 3-0 are returned zeroed. ^a

Table B-2 AACISDATAIN slot description (continued)

3	Left channel PCM capture data	20	This slot contains the left channel ADC data. The signal digitized is selected through register 1A, that defaults to 0000. This selects the MIC input. This is a 20-bit slot, where the PCM data returned is MSB first and the last remaining bits are 0.
4	PCM record right channel	20	This slot contains the right channel ADC data. The signal digitized is selected through register 1A, that defaults to 0000. This selects the MIC input. This is a 20-bit slot, where the PCM data returned is MSB first and the last two remaining bits are 0.
5	Modem 1 ADC	20	This slot contains data from the Modem Line 1 ADC output data.
6	Microphone record data	20	This slot contains data from the microphone record ADC.
7-9	Reserved	20	These slots are reserved. However, if data is required from the slot the receive you can configure the FIFOs to accept the data sent.
10	Modem 2 ADC	20	This slot contains data from the Modem Line 2 ADC.
11	Headset ADC	20	This slot contains data from Headset ADC.
12	I/O status	20	This slot contains I/O status data.

a. The Cirrus device implements bus-keeper logic for its 16-bit registers. Therefore no PrimeCell AACI software dependencies must exist on the value of undefined register bits returned on a read. For a list of undefined bits in the register map see the Cirrus CS4299 device specification.

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